

BFS Traversal

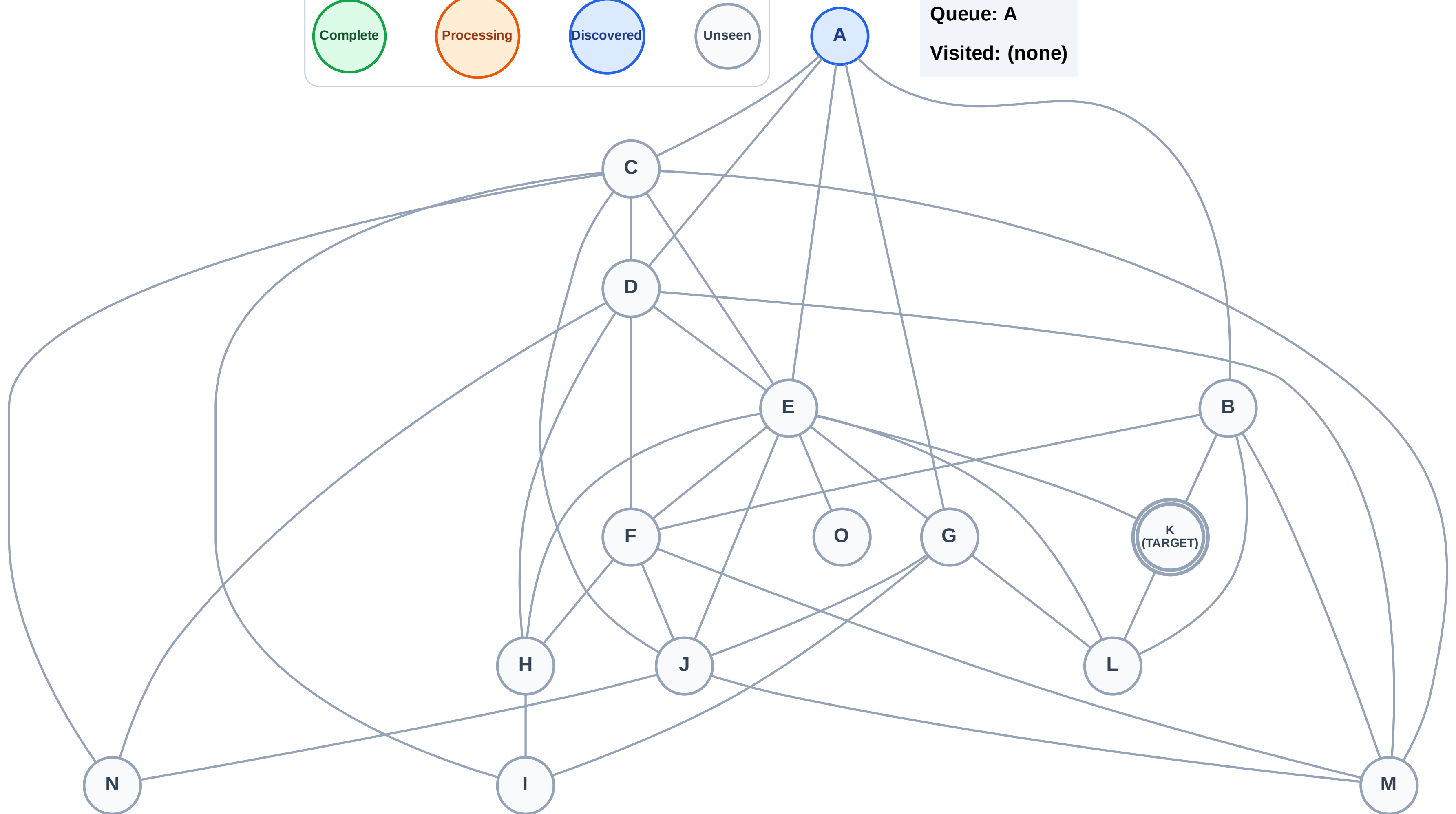
Step 1: Start at A (Seeking target node K)

Legend



Queue: A

Visited: (none)



BFS Traversal

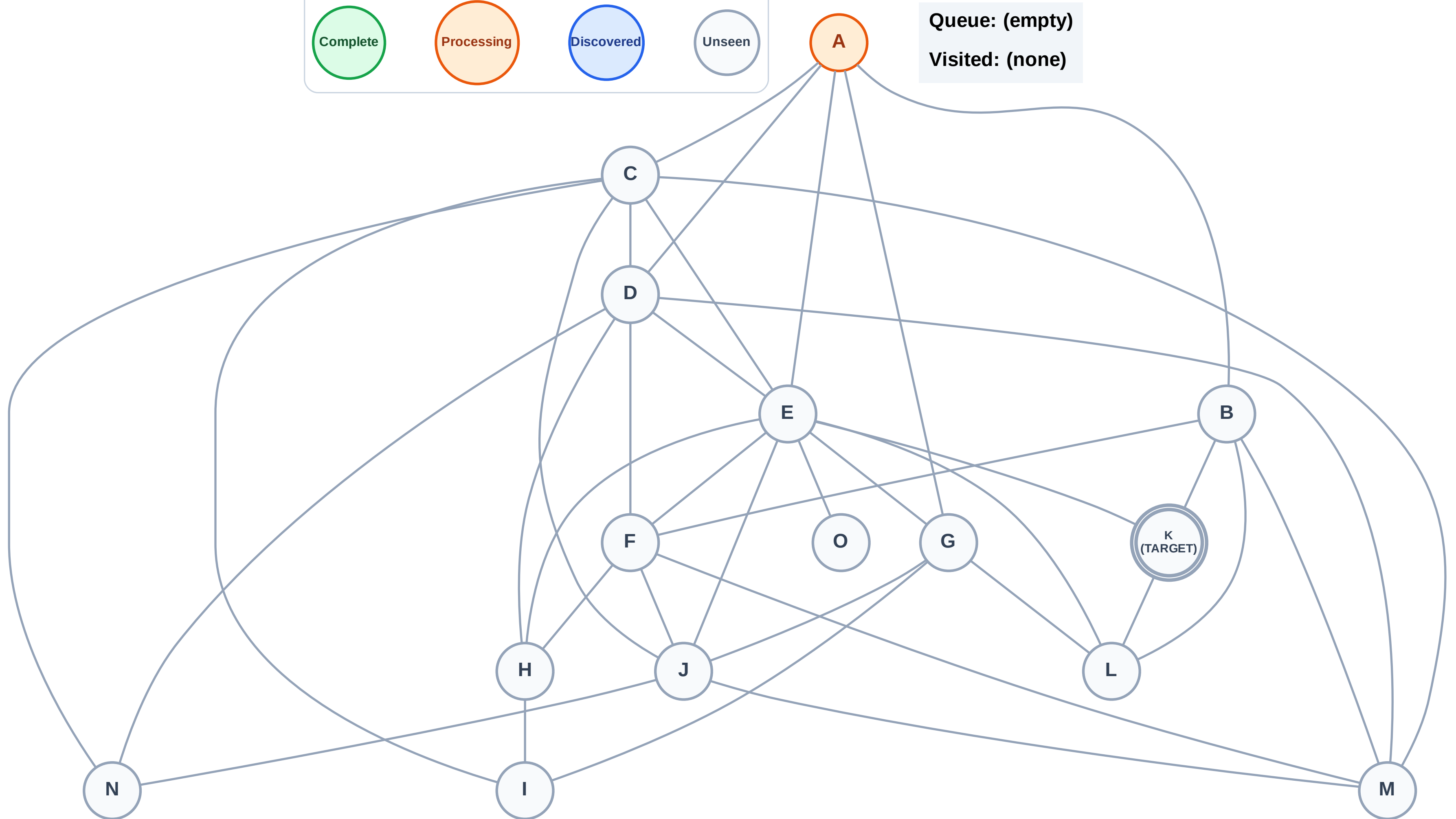
Step 2: Process node A

Legend



Queue: (empty)

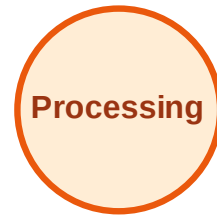
Visited: (none)



BFS Traversal

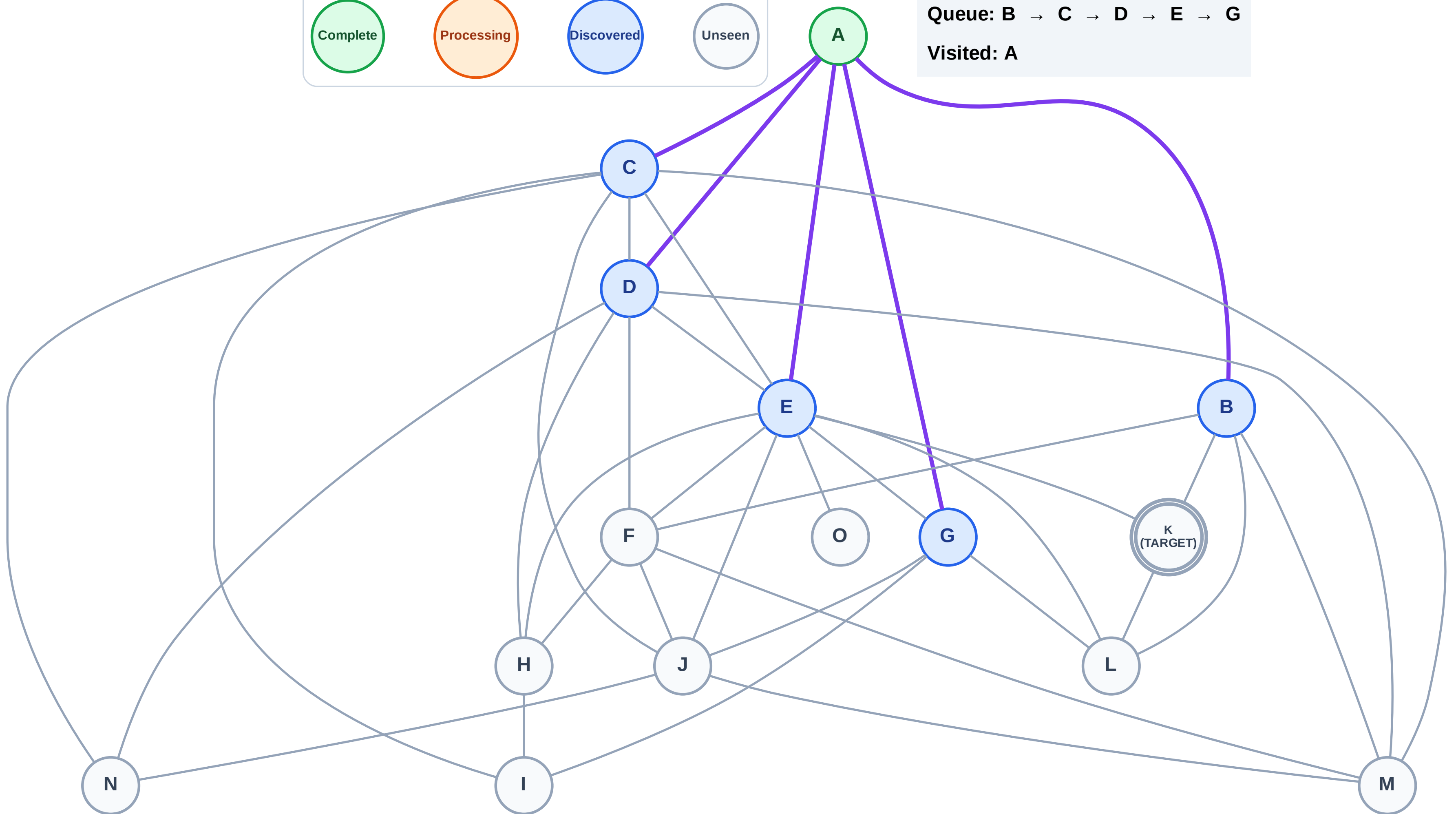
Step 3: Discover B, C, D, E, G from A

Legend



Queue: B → C → D → E → G

Visited: A



BFS Traversal

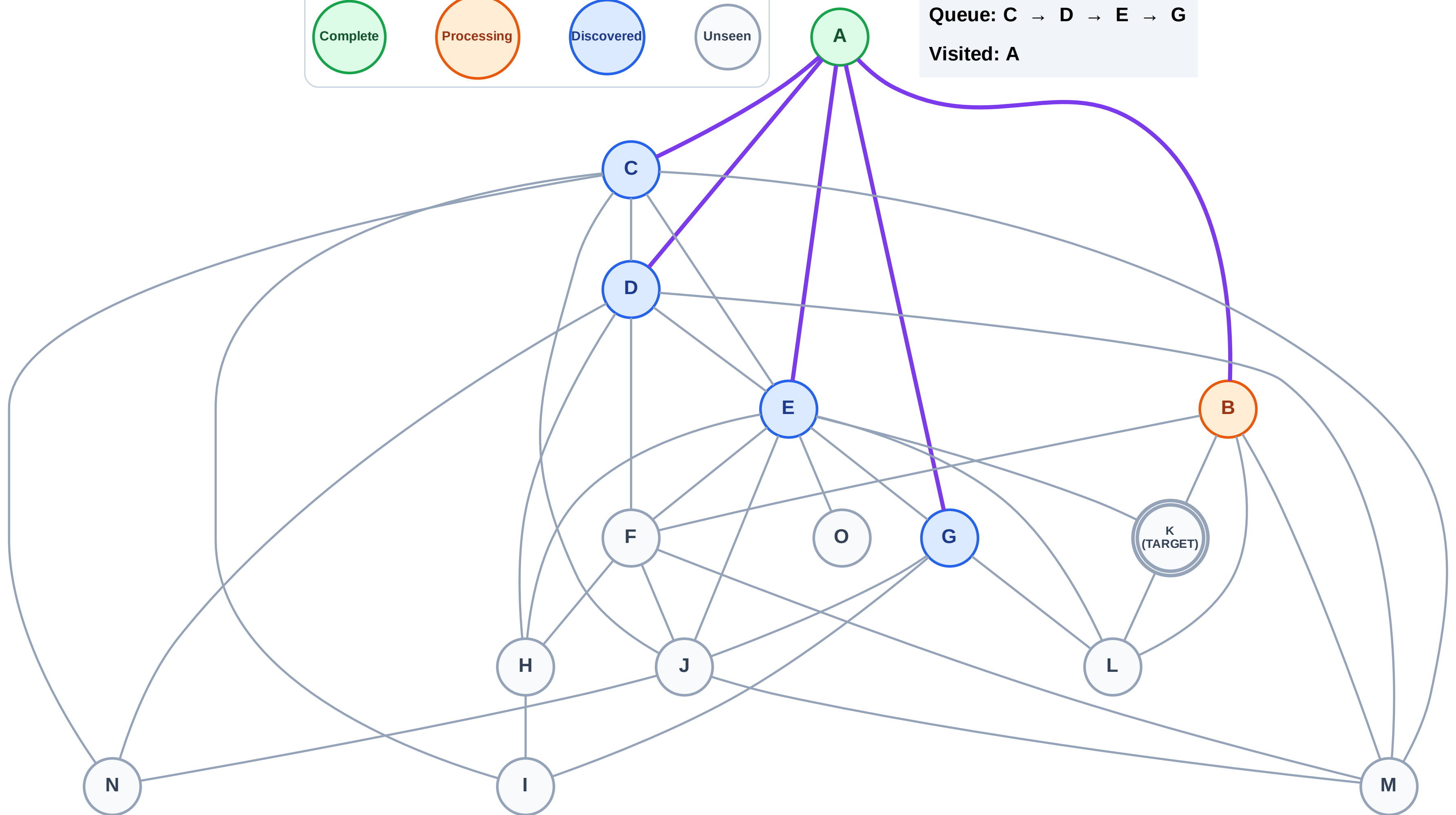
Step 4: Process node B

Legend



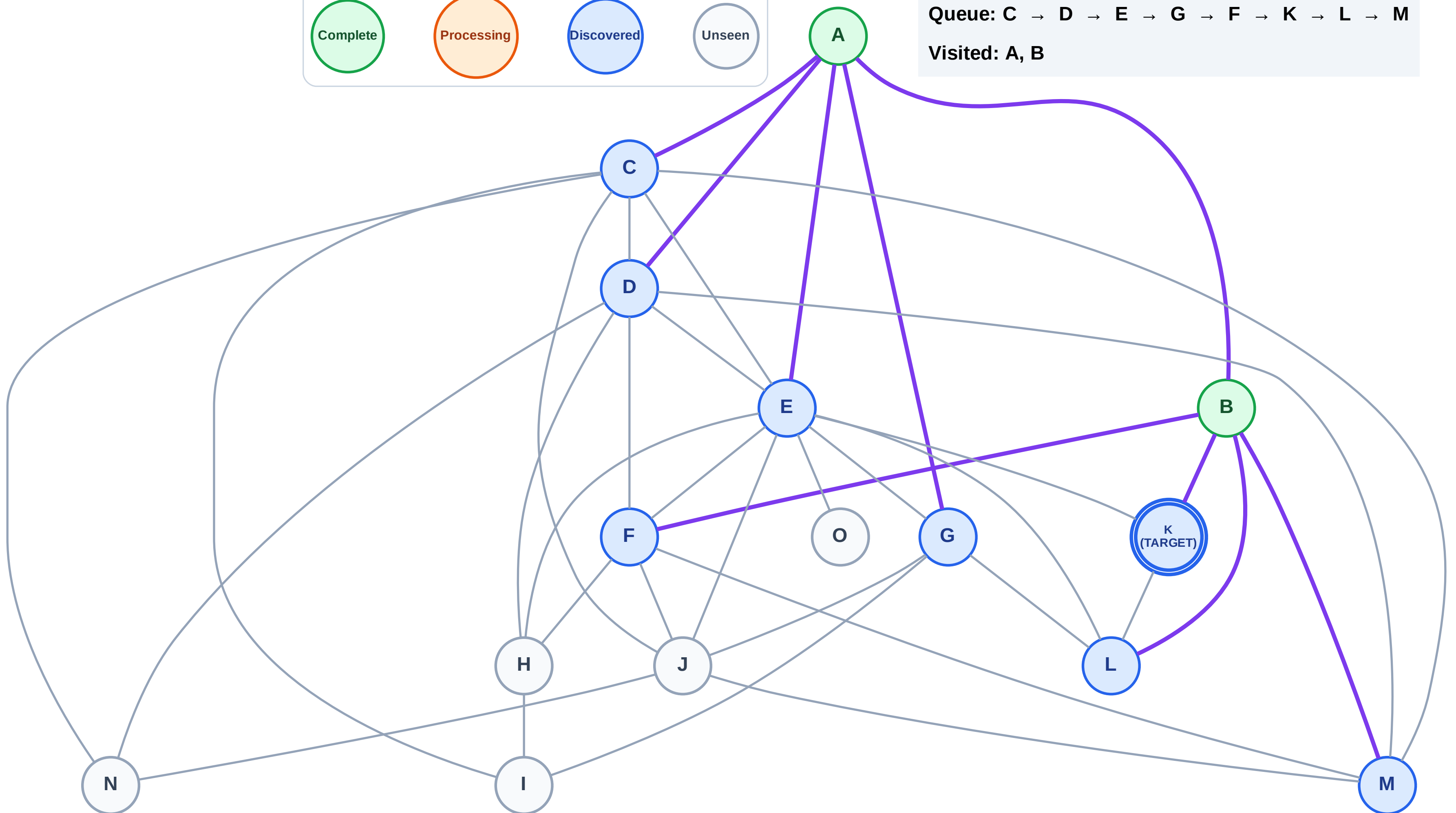
Queue: C → D → E → G

Visited: A



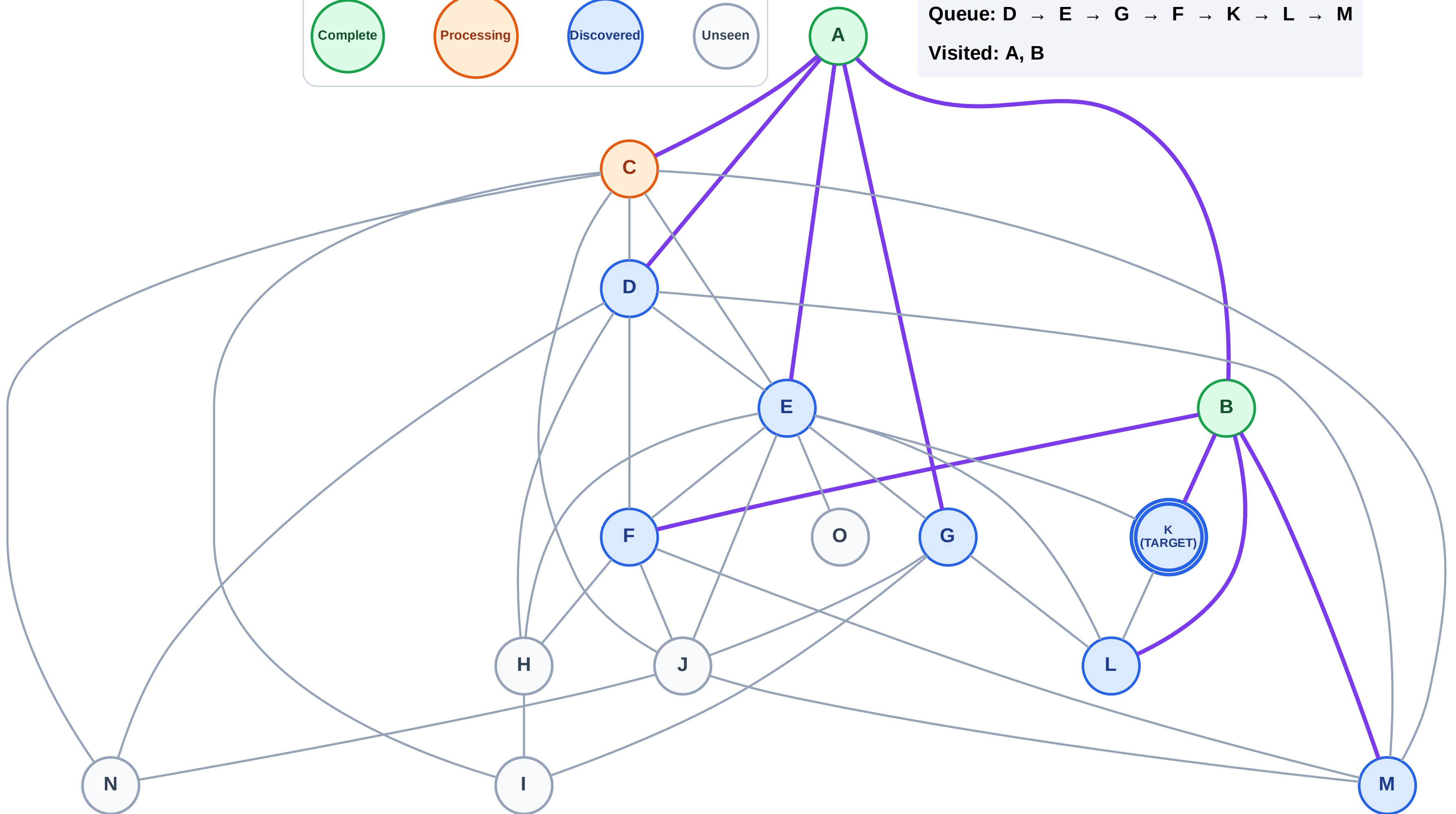
Step 5: Discover F, K, L, M from B

Step 5: Discover F, K, L, M from B



Step 6: Process node C

Step 6: Process node C



BFS Traversal

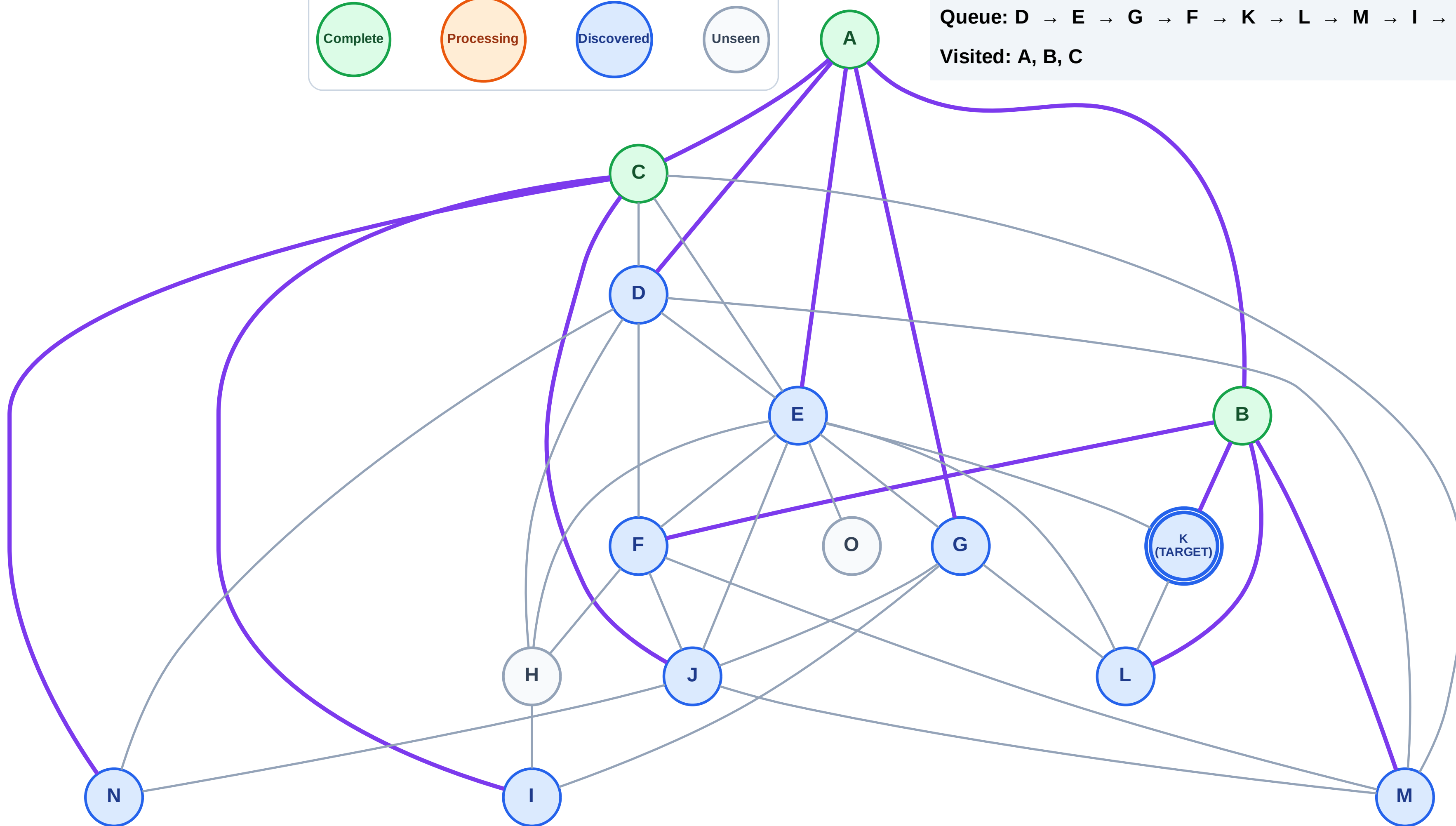
Step 7: Discover I, J, N from C

Legend



Queue: D → E → G → F → K → L → M → I → J → N

Visited: A, B, C



BFS Traversal

Step 8: Process node D

Legend

Complete

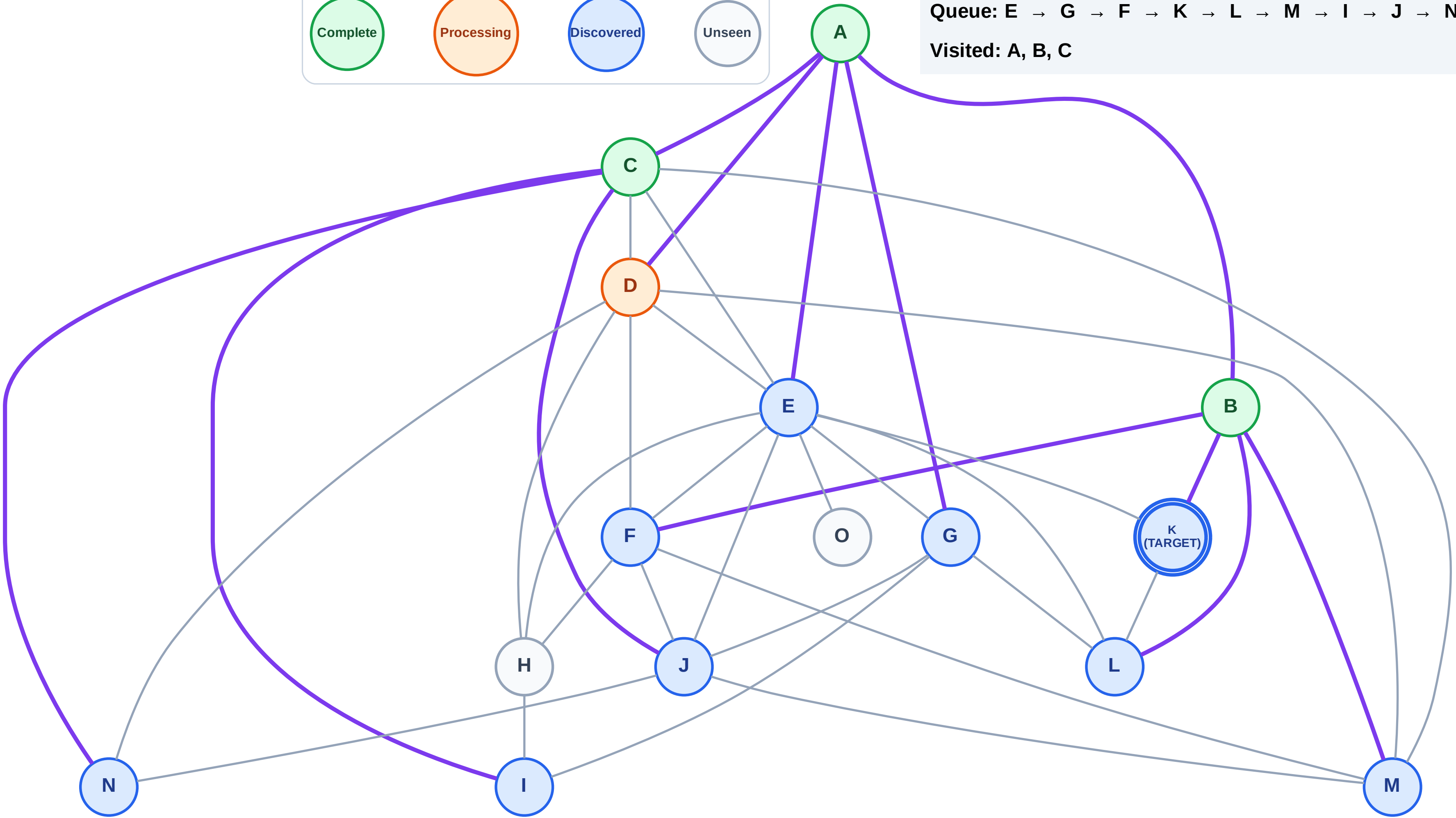
Processing

Discovered

Unseen

Queue: E → G → F → K → L → M → I → J → N

Visited: A, B, C



BFS Traversal

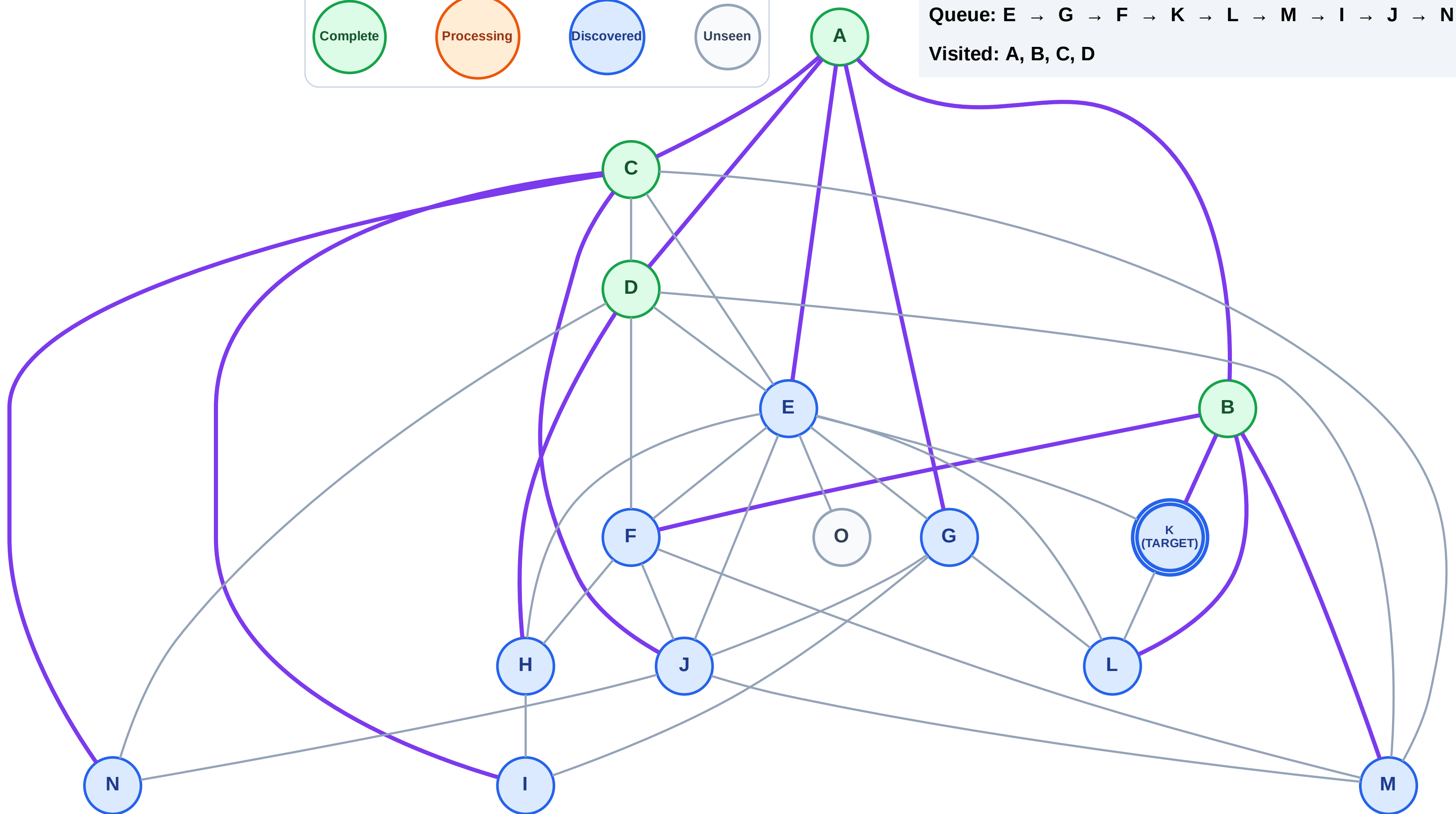
Step 9: Discover H from D

Legend



Queue: E → G → F → K → L → M → I → J → N → H

Visited: A, B, C, D



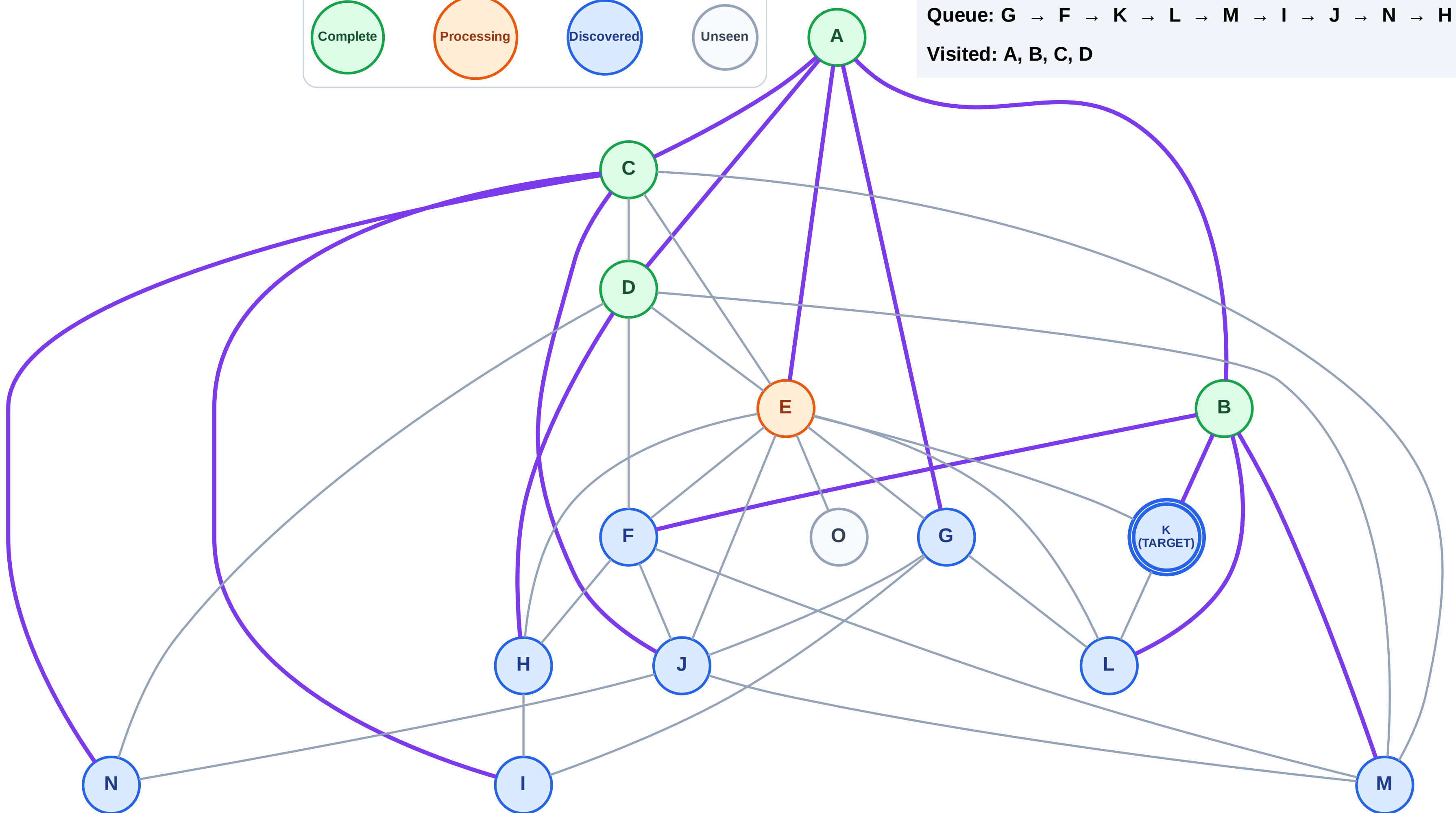
BFS Traversal

Step 10: Process node E

Legend



Queue: G → F → K → L → M → I → J → N → H
Visited: A, B, C, D



BFS Traversal

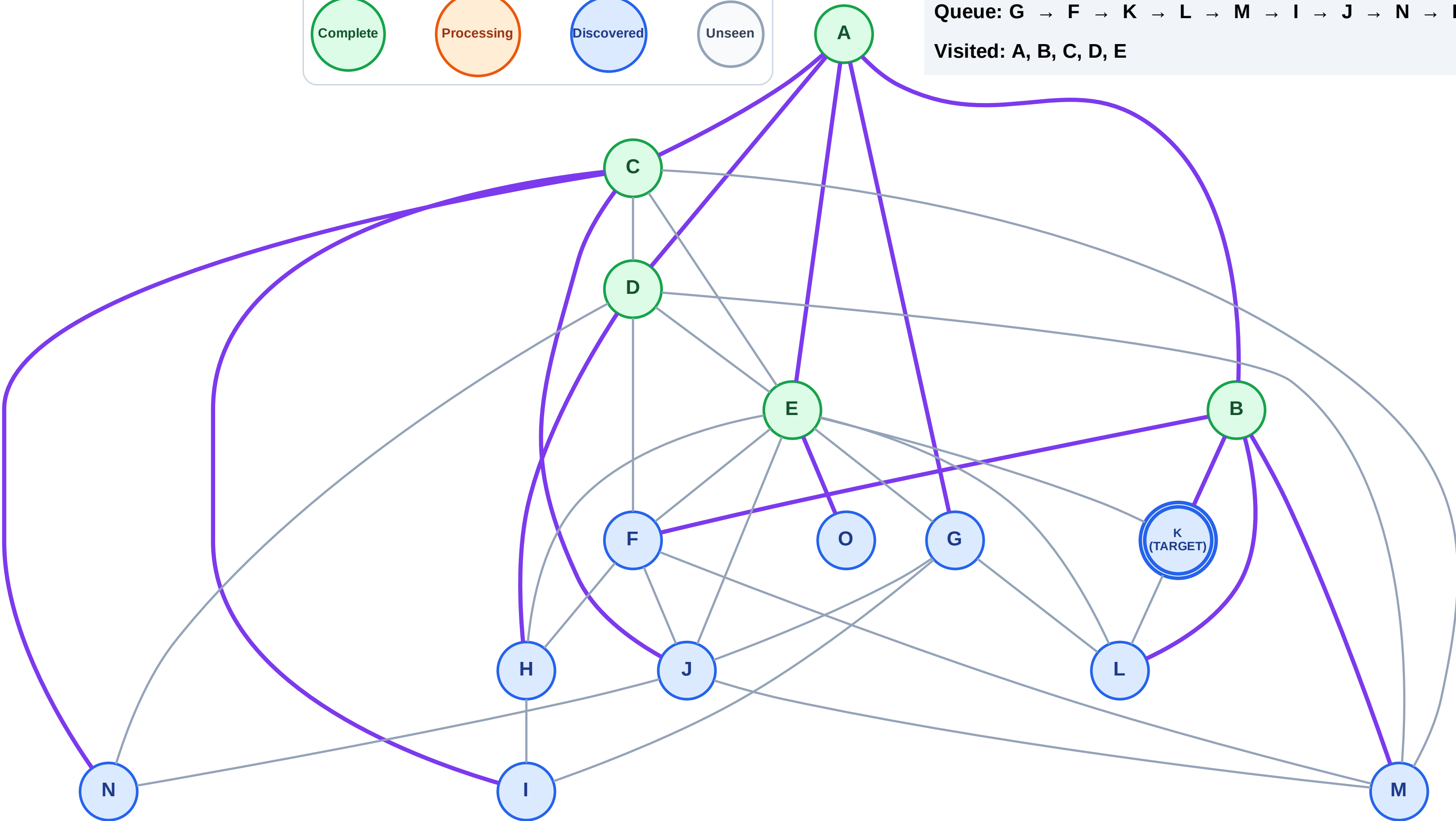
Step 11: Discover O from E

Legend



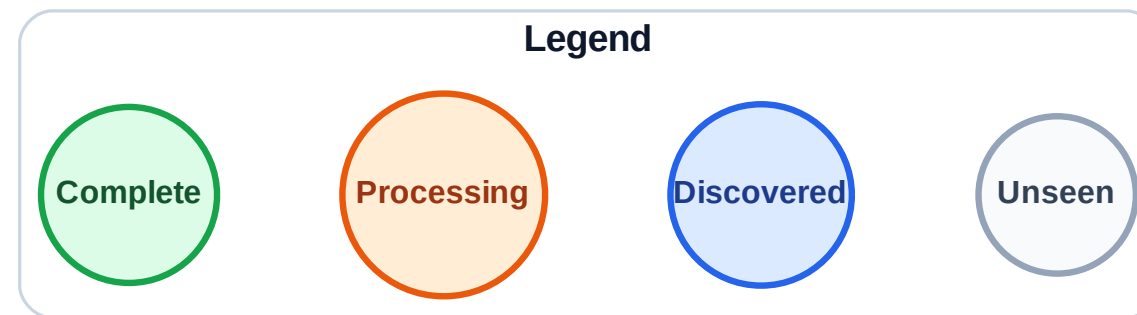
Queue: G → F → K → L → M → I → J → N → H → O

Visited: A, B, C, D, E



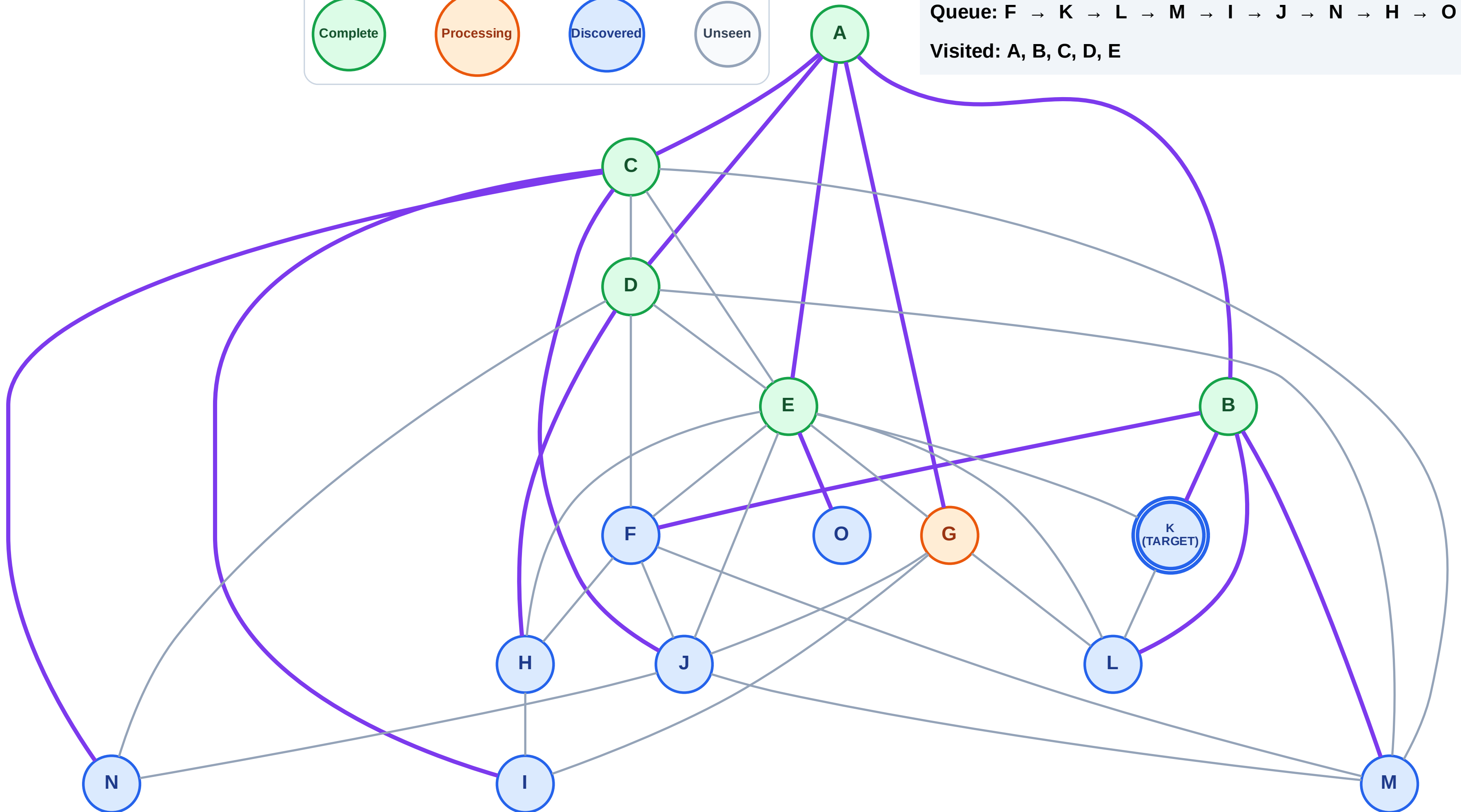
BFS Traversal

Step 12: Process node G



Queue: F → K → L → M → I → J → N → H → O

Visited: A, B, C, D, E



BFS Traversal

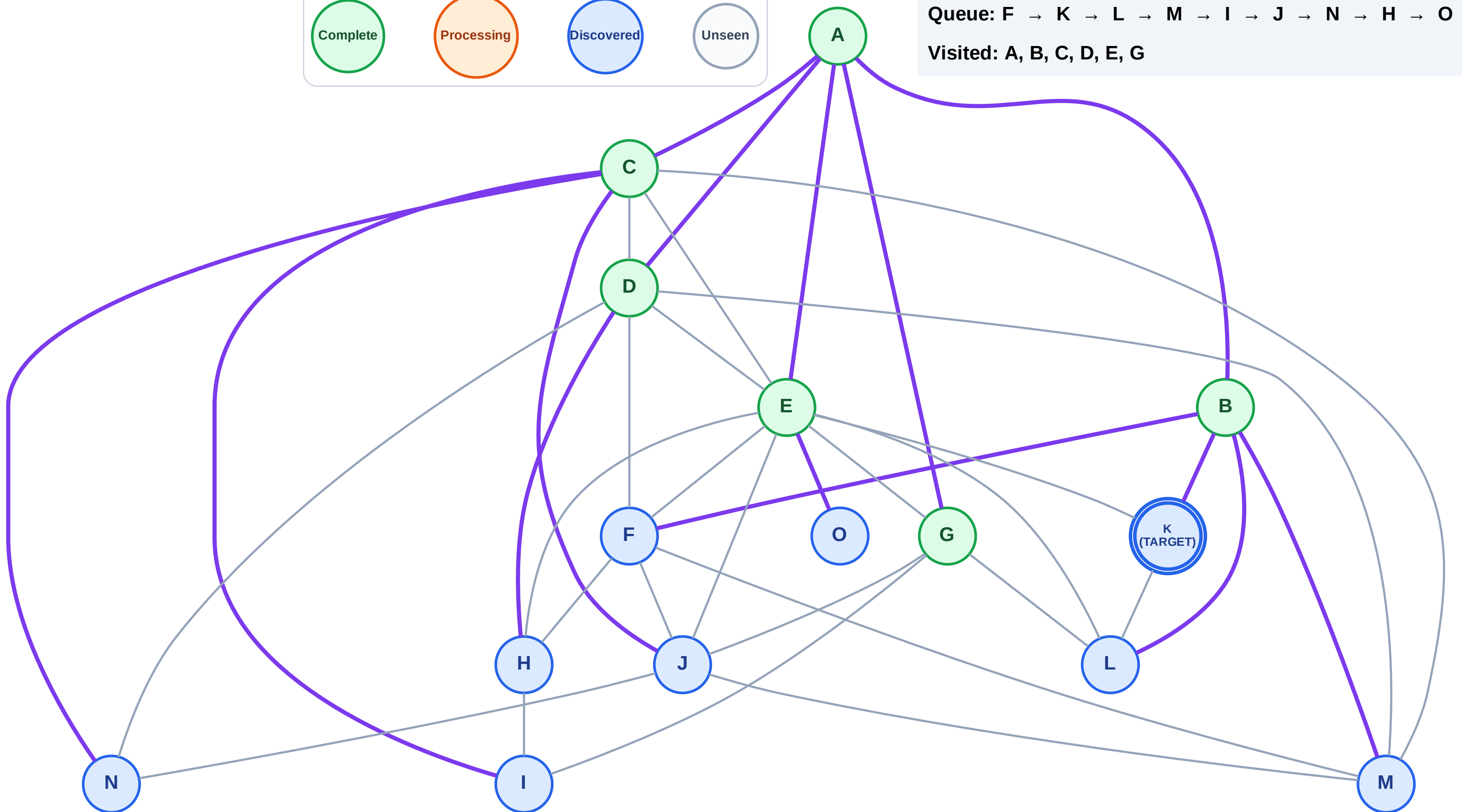
Step 13: No new neighbors from G

Legend



Queue: F → K → L → M → I → J → N → H → O

Visited: A, B, C, D, E, G



BFS Traversal

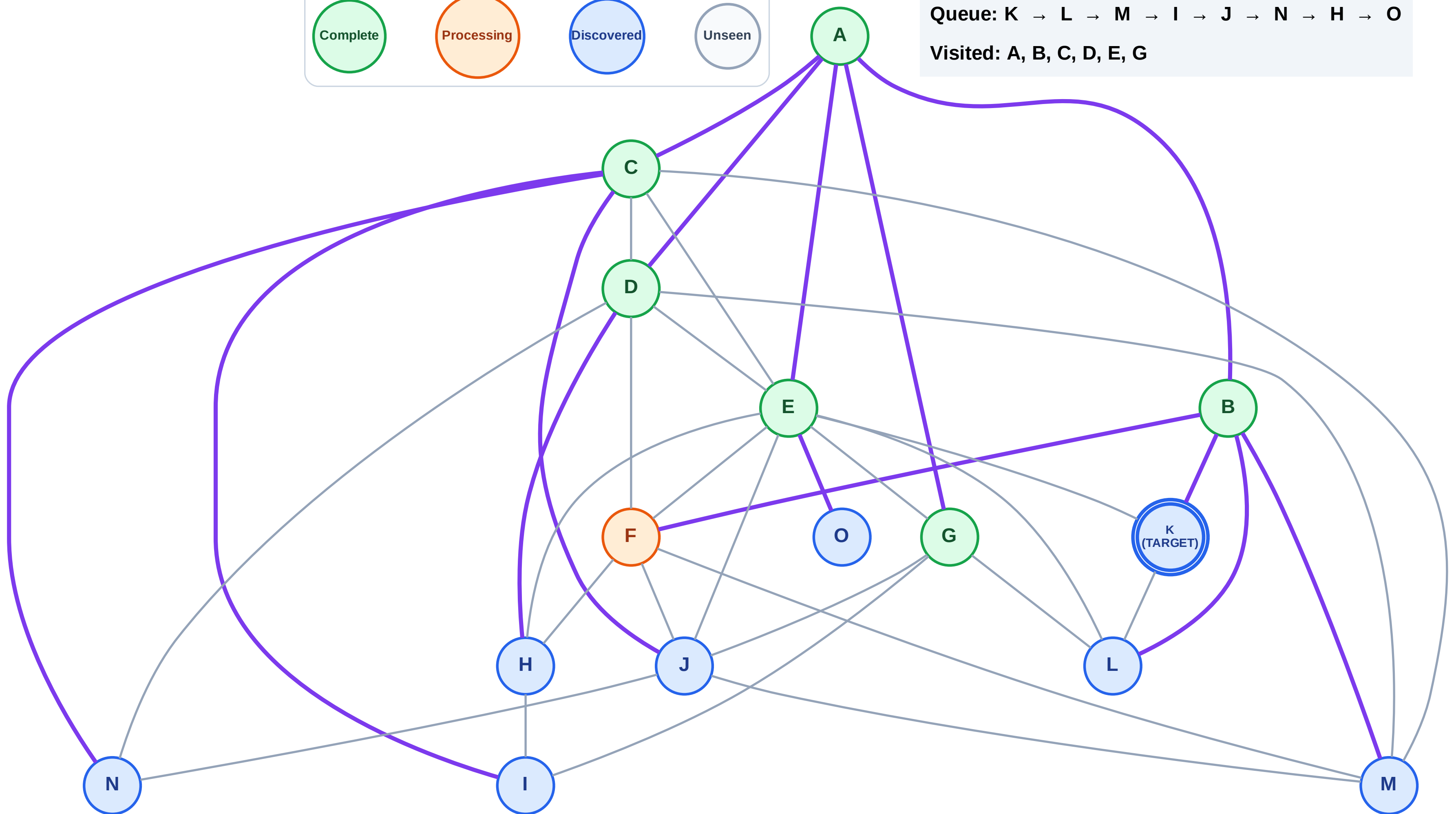
Step 14: Process node F

Legend



Queue: K → L → M → I → J → N → H → O

Visited: A, B, C, D, E, G



BFS Traversal

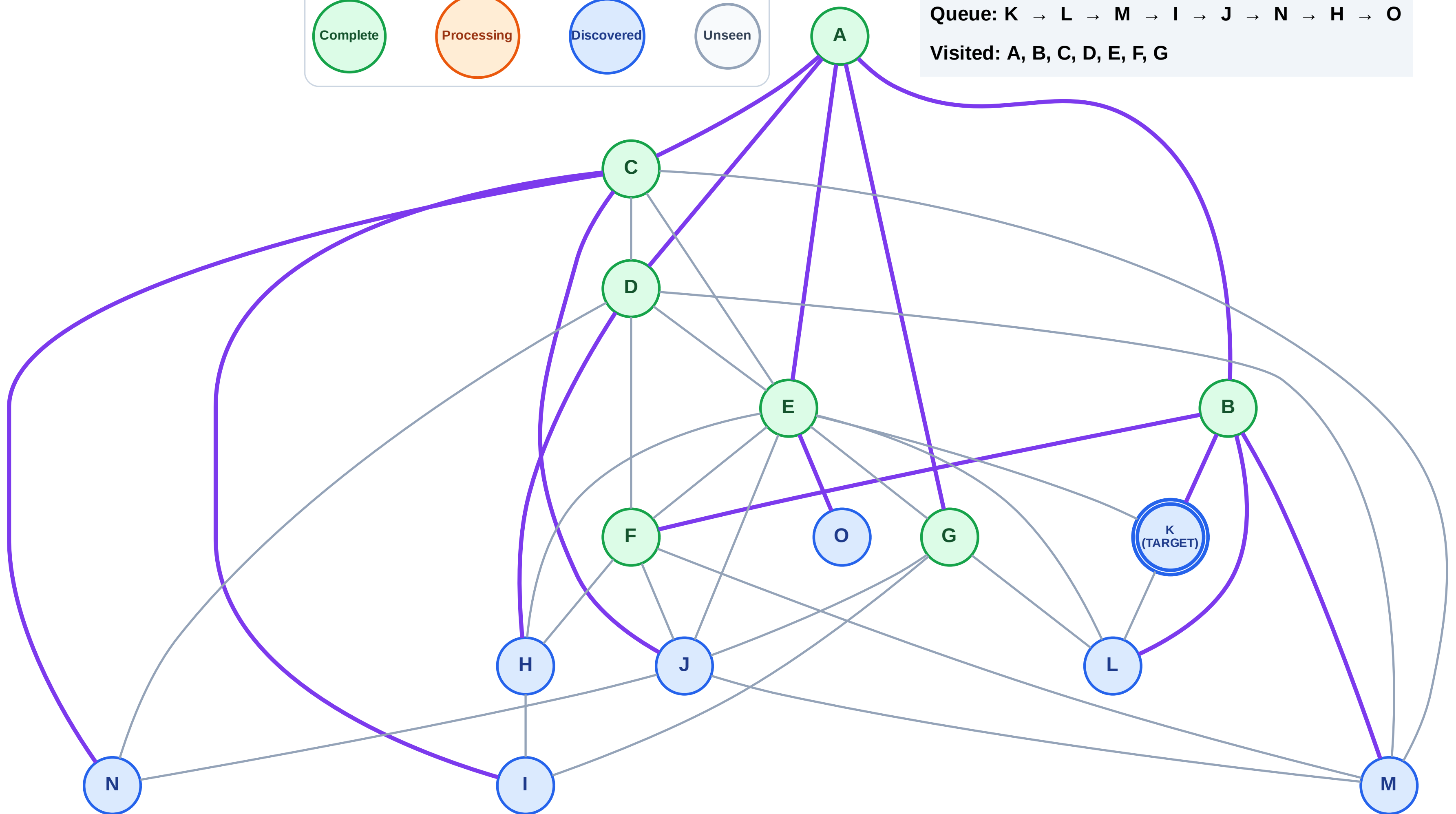
Step 15: No new neighbors from F

Legend



Queue: K → L → M → I → J → N → H → O

Visited: A, B, C, D, E, F, G



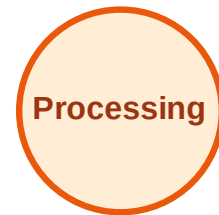
BFS Traversal

Step 16: Process node K

Legend



Complete



Processing



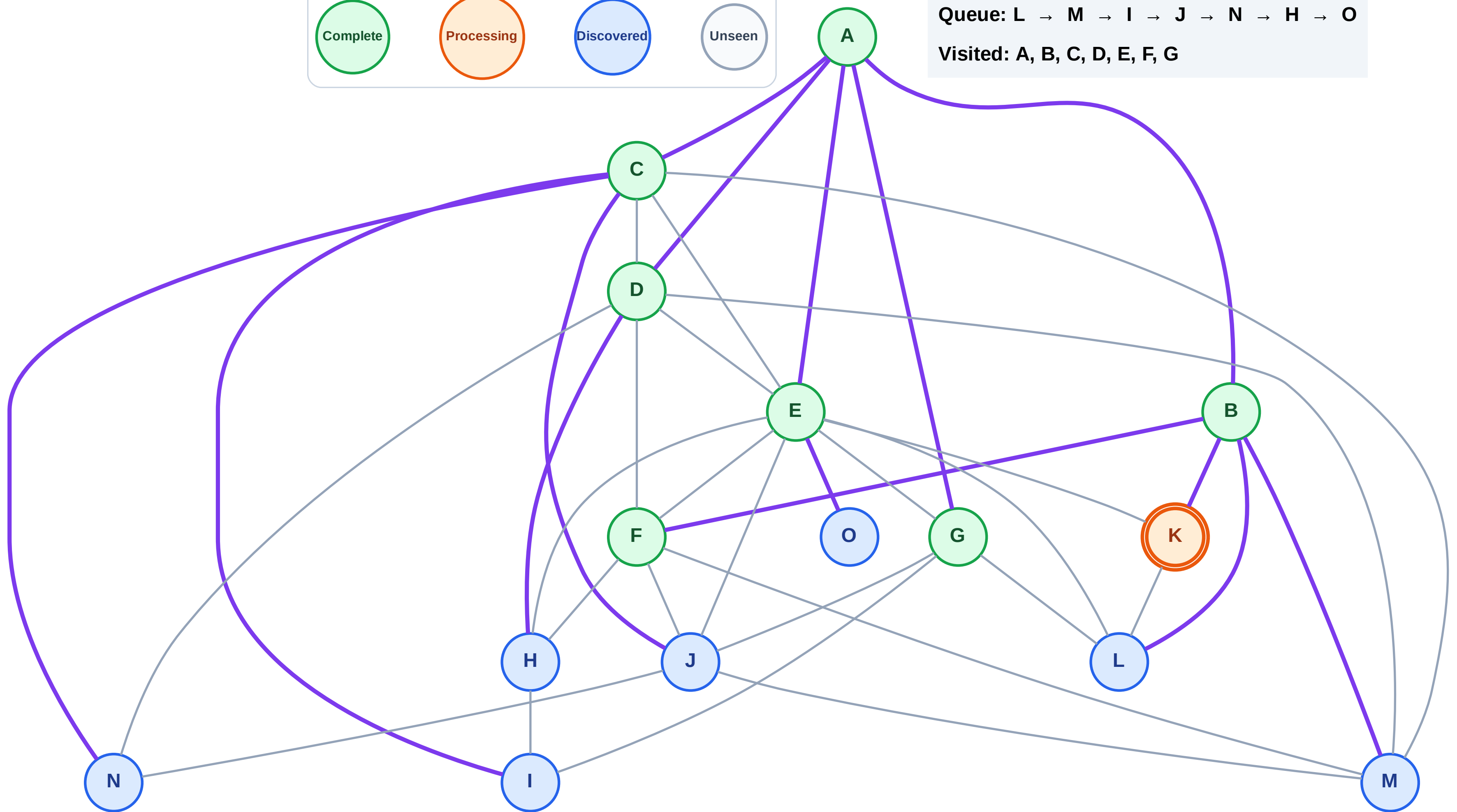
Discovered



Unseen

Queue: L → M → I → J → N → H → O

Visited: A, B, C, D, E, F, G



BFS Traversal

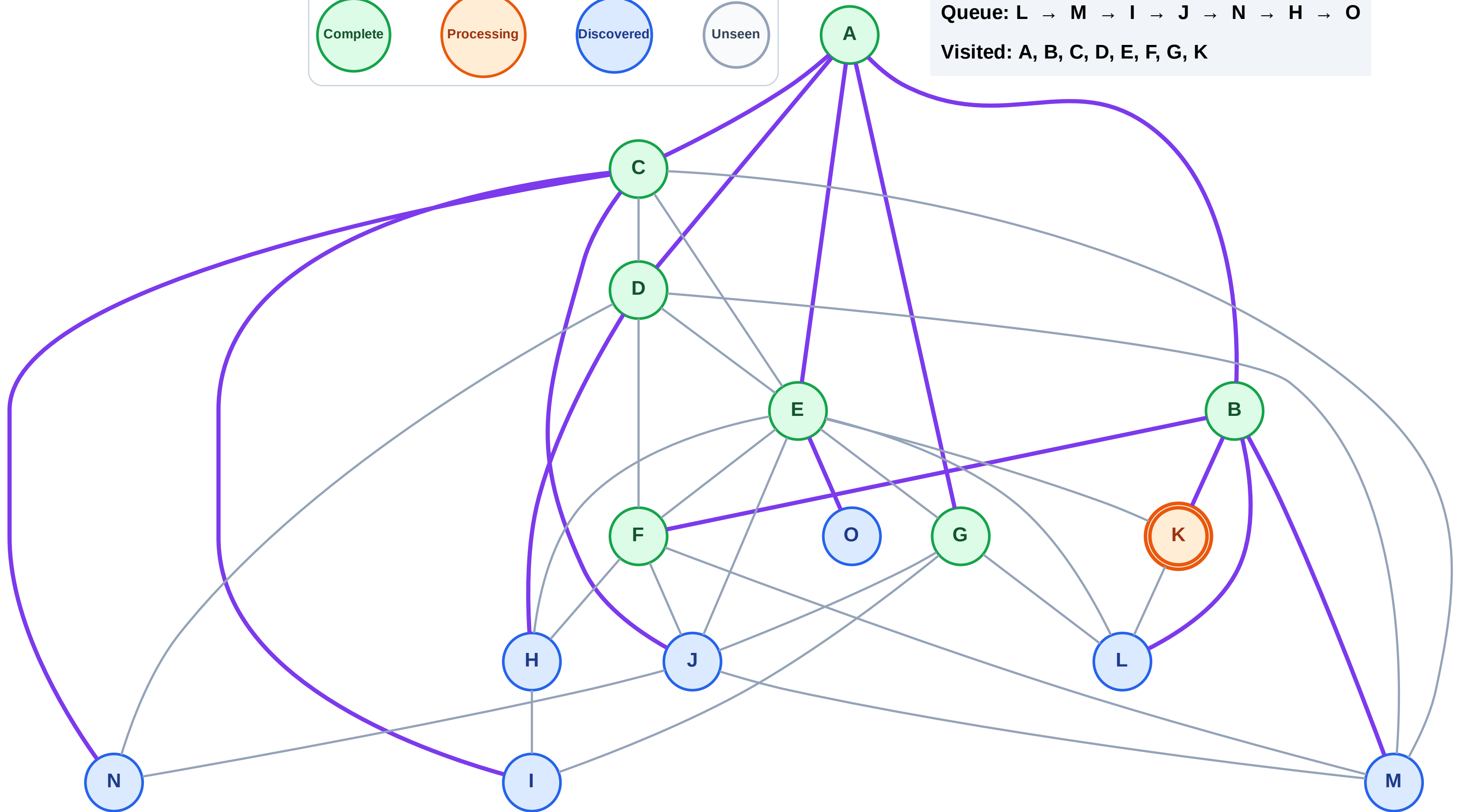
Step 17: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend



Queue: L → M → I → J → N → H → O

Visited: A, B, C, D, E, F, G, K



DFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



Complete



Processing



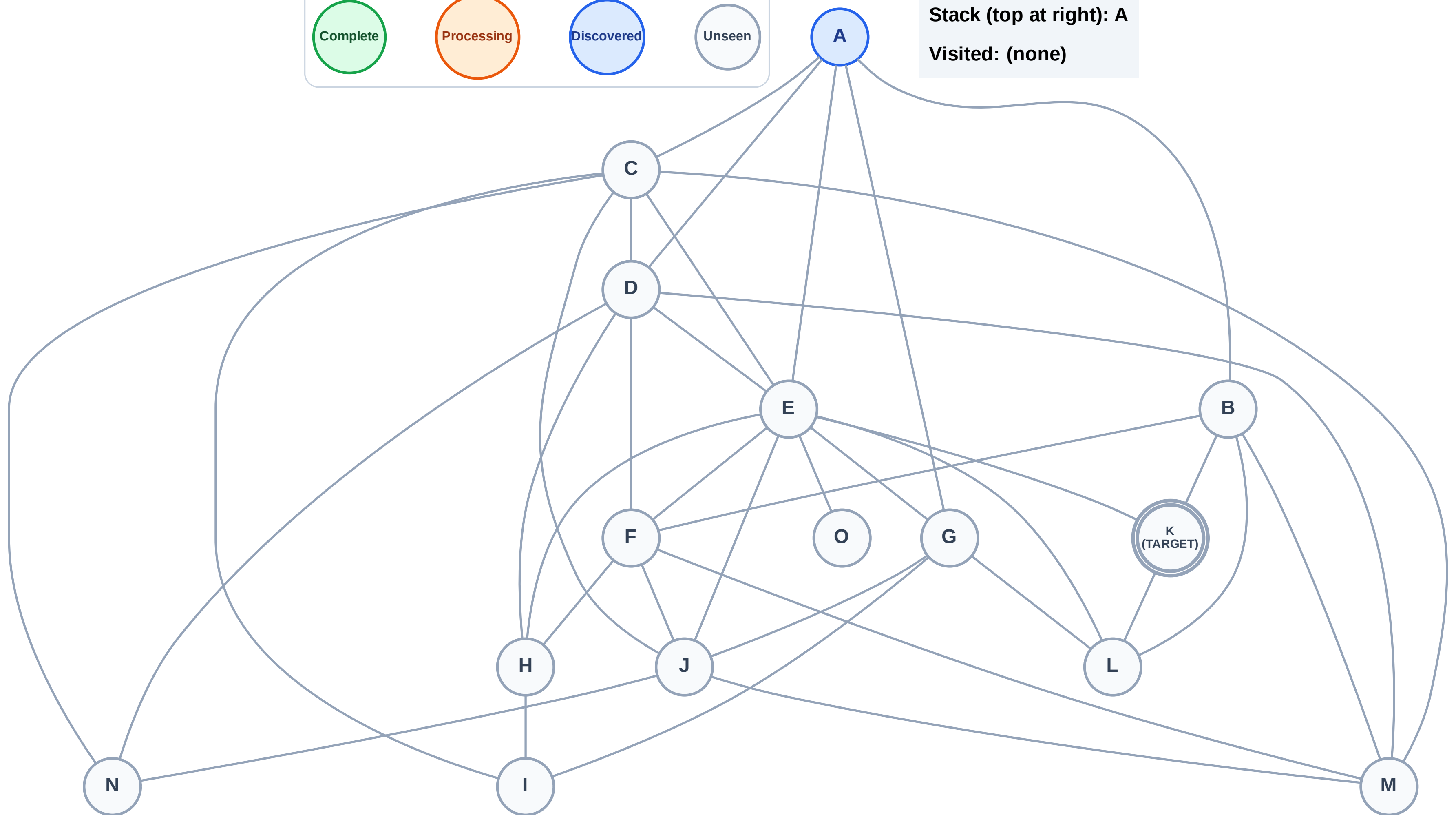
Discovered



Unseen

Stack (top at right): A

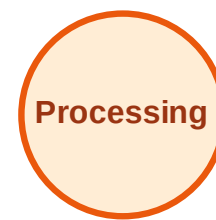
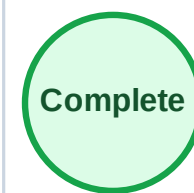
Visited: (none)



DFS Traversal

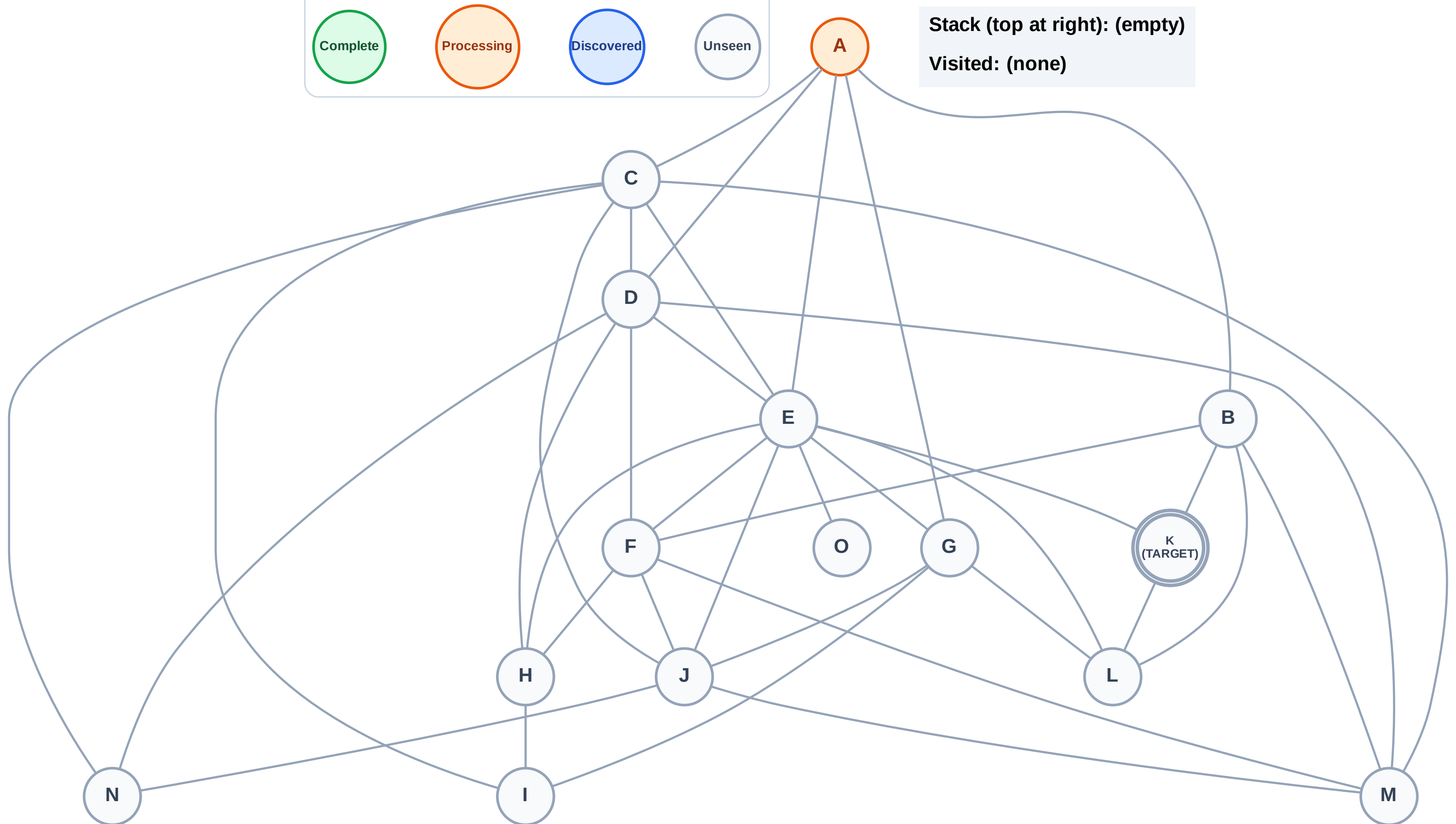
Step 2: Process node A

Legend



Stack (top at right): (empty)

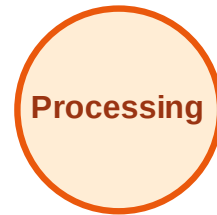
Visited: (none)



DFS Traversal

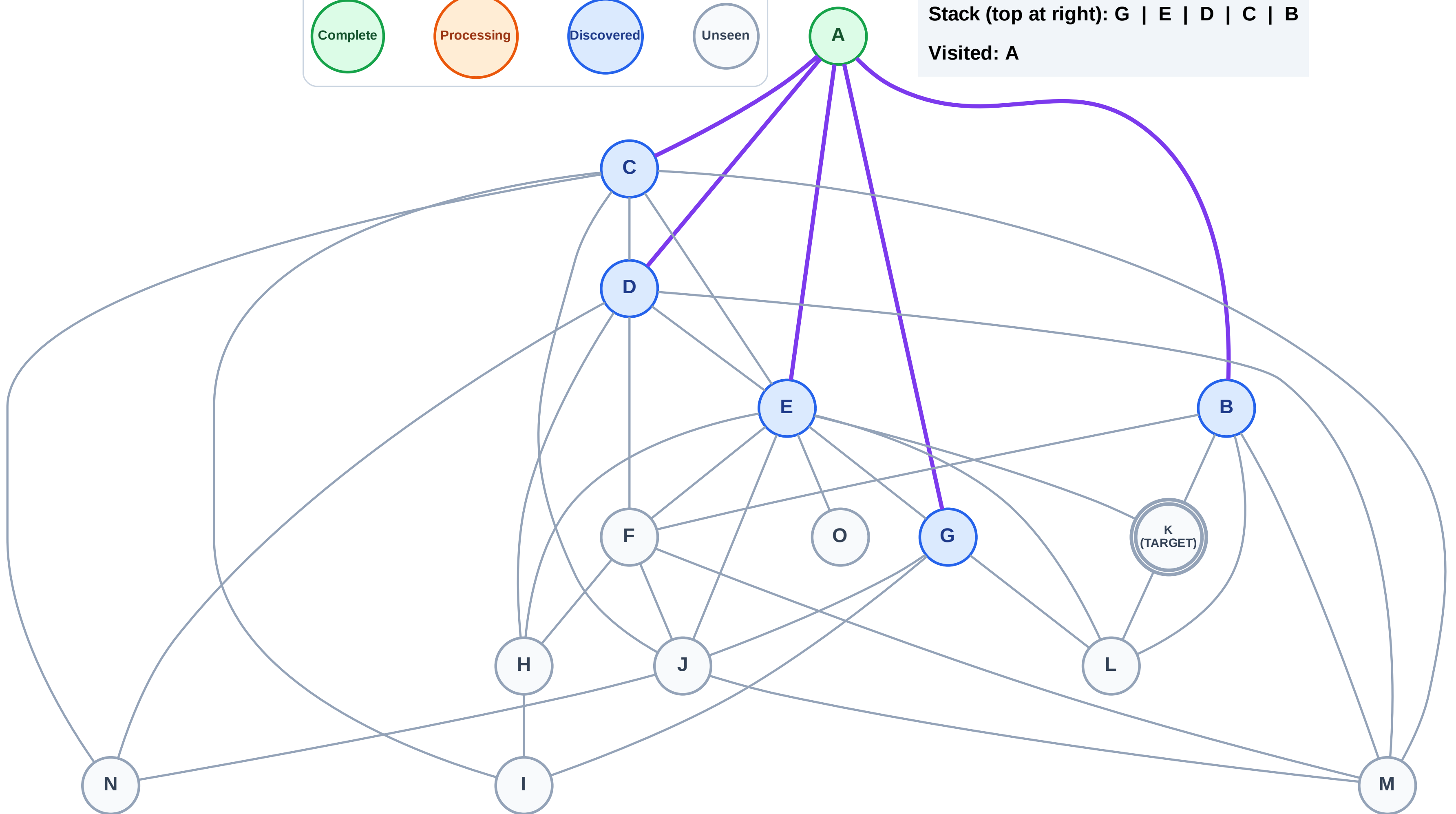
Step 3: Discover G, E, D, C, B from A

Legend



Stack (top at right): G | E | D | C | B

Visited: A



DFS Traversal

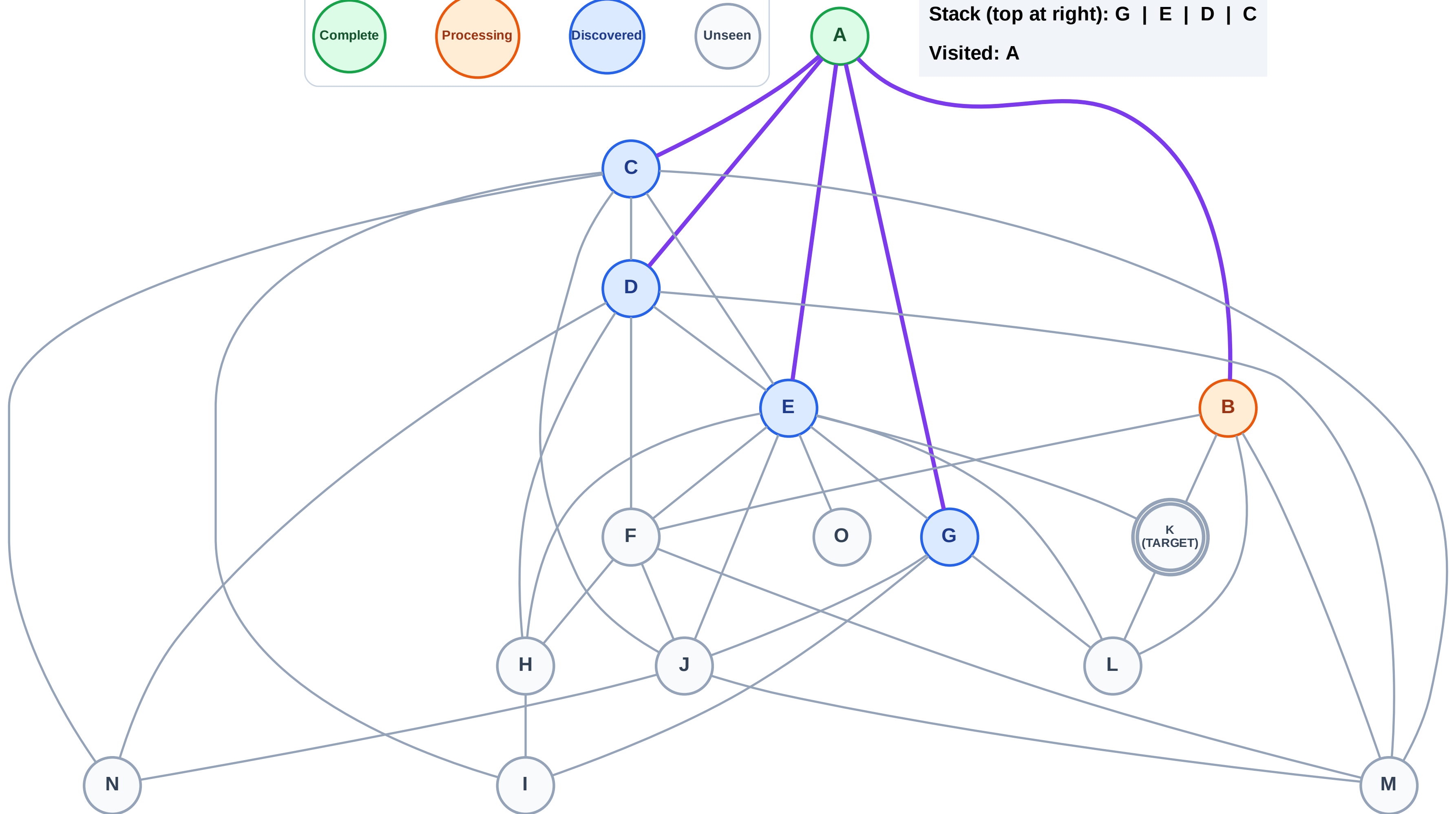
Step 4: Process node B

Legend



Stack (top at right): G | E | D | C

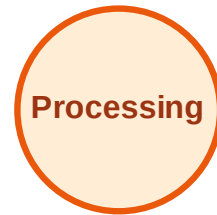
Visited: A



DFS Traversal

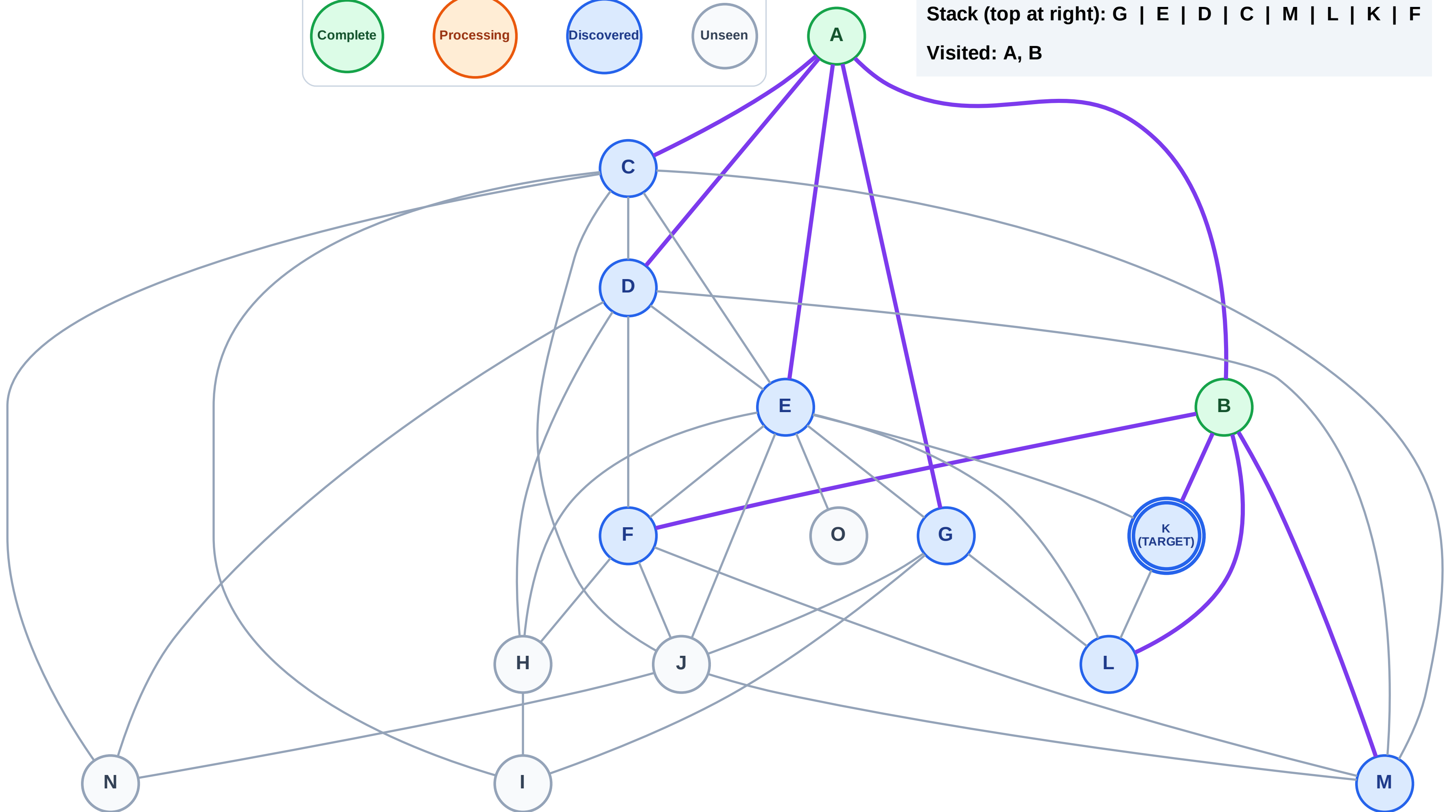
Step 5: Discover M, L, K, F from B

Legend



Stack (top at right): G | E | D | C | M | L | K | F

Visited: A, B



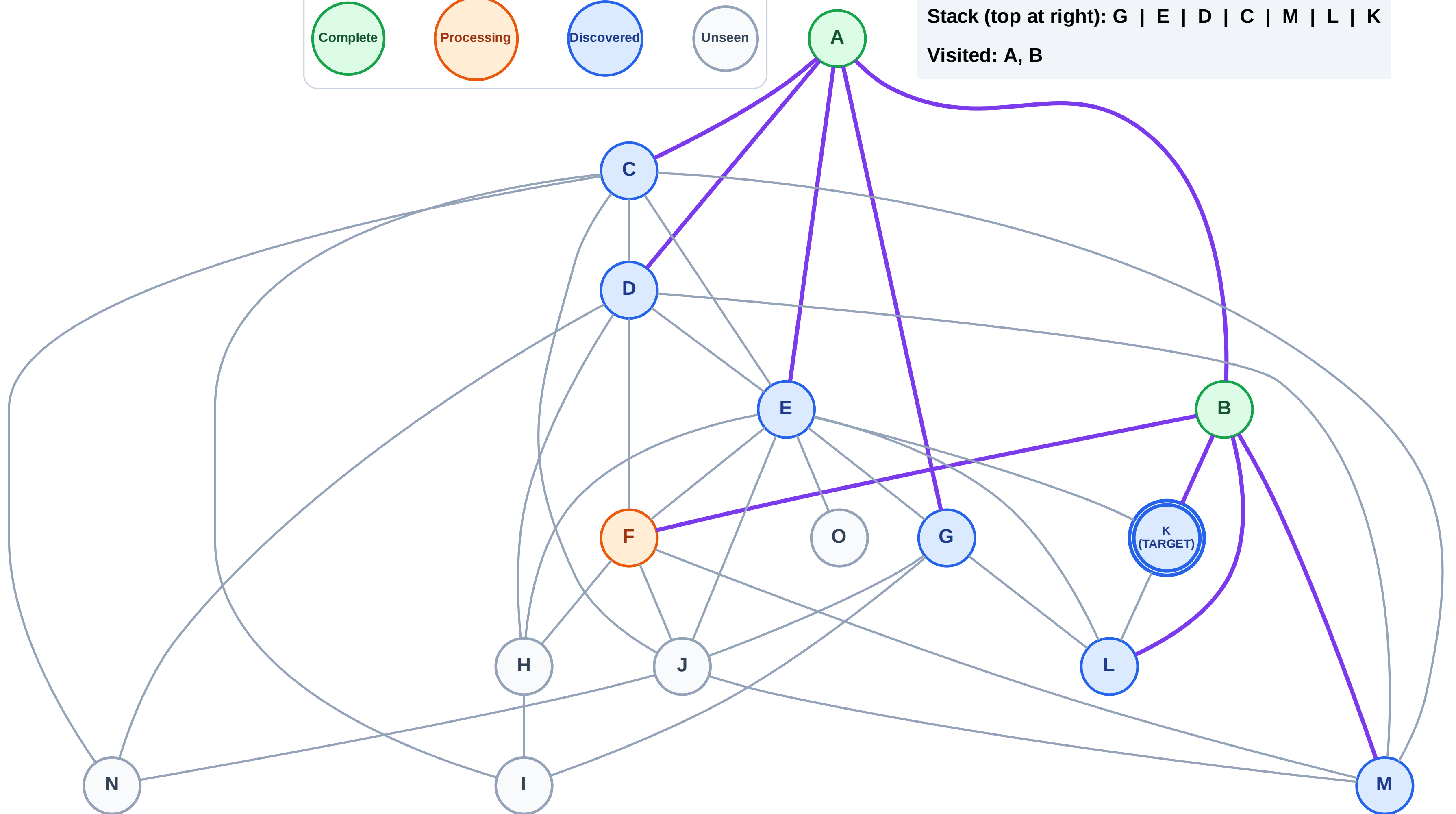
DFS Traversal

Step 6: Process node F

Legend



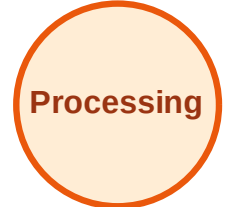
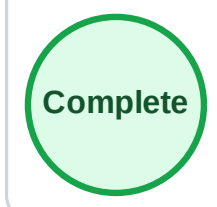
Stack (top at right): G | E | D | C | M | L | K
Visited: A, B



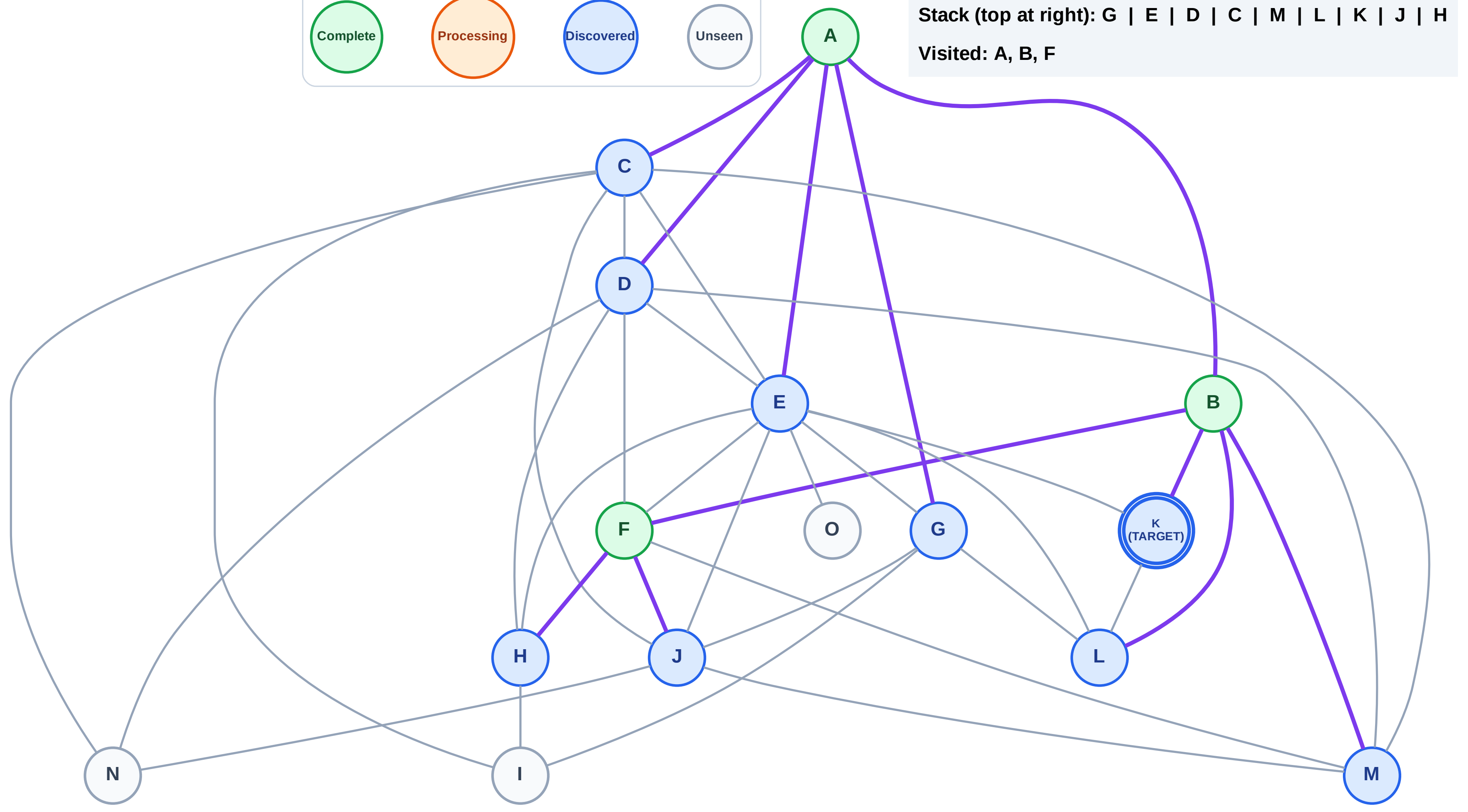
DFS Traversal

Step 7: Discover J, H from F

Legend



Stack (top at right): G | E | D | C | M | L | K | J | H
Visited: A, B, F



DFS Traversal

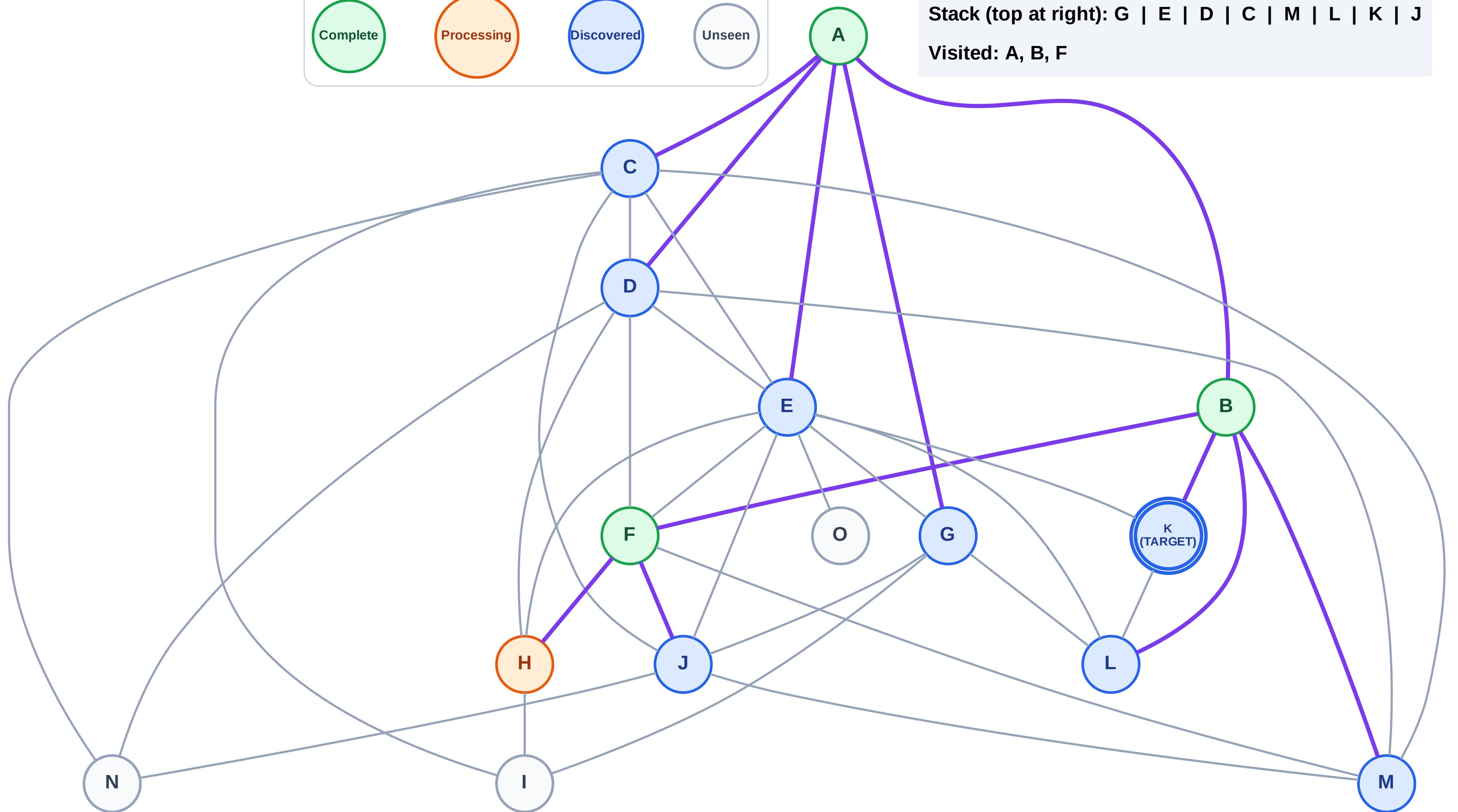
Step 8: Process node H

Legend



Stack (top at right): G | E | D | C | M | L | K | J

Visited: A, B, F



Step 9: Discover I from H

Step 9: Discover I from H

Legend

Complete

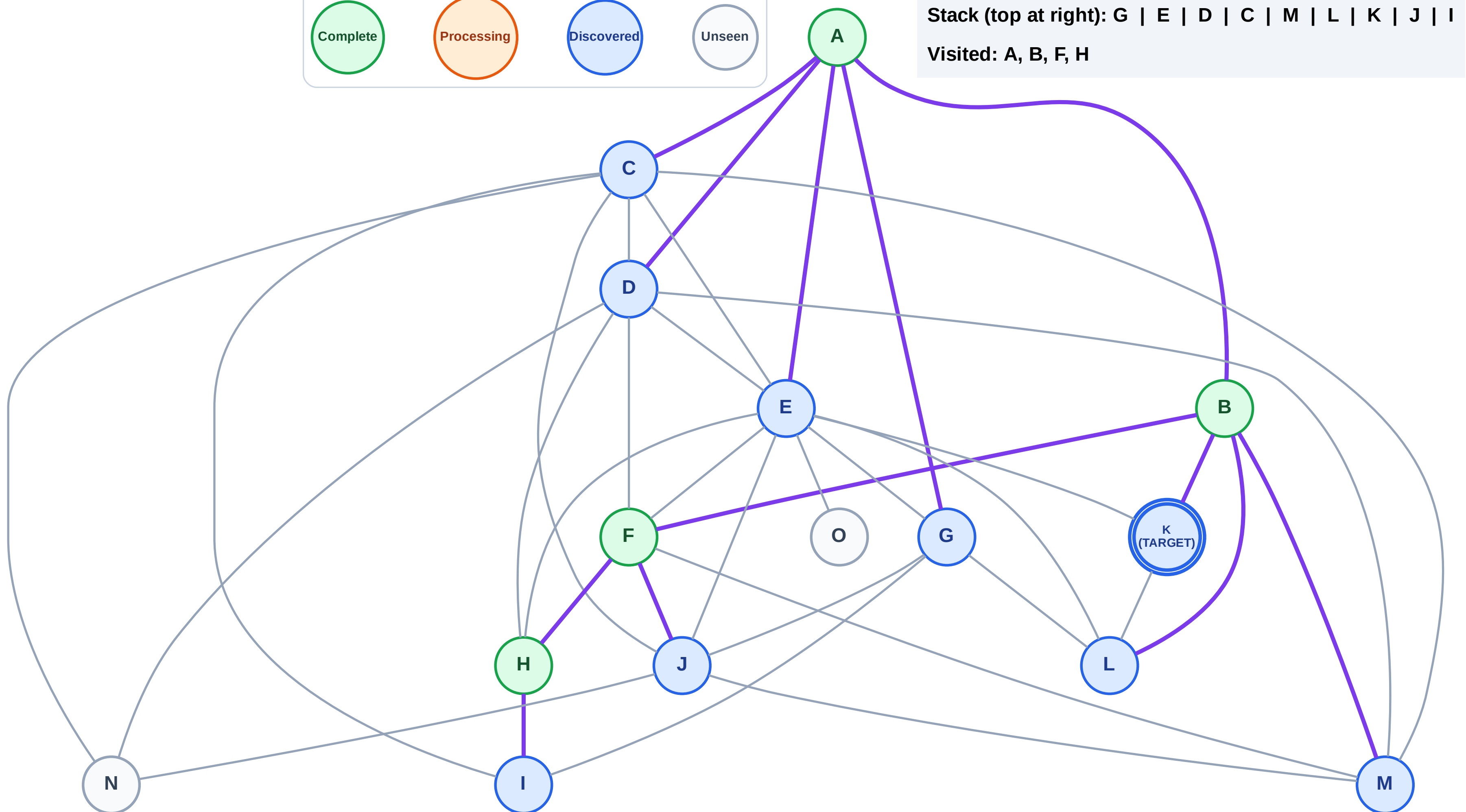
Processing

Discovered

Unseen

Stack (top at right): G | E | D | C | M | L | K | J | I

Visited: A, B, F, H



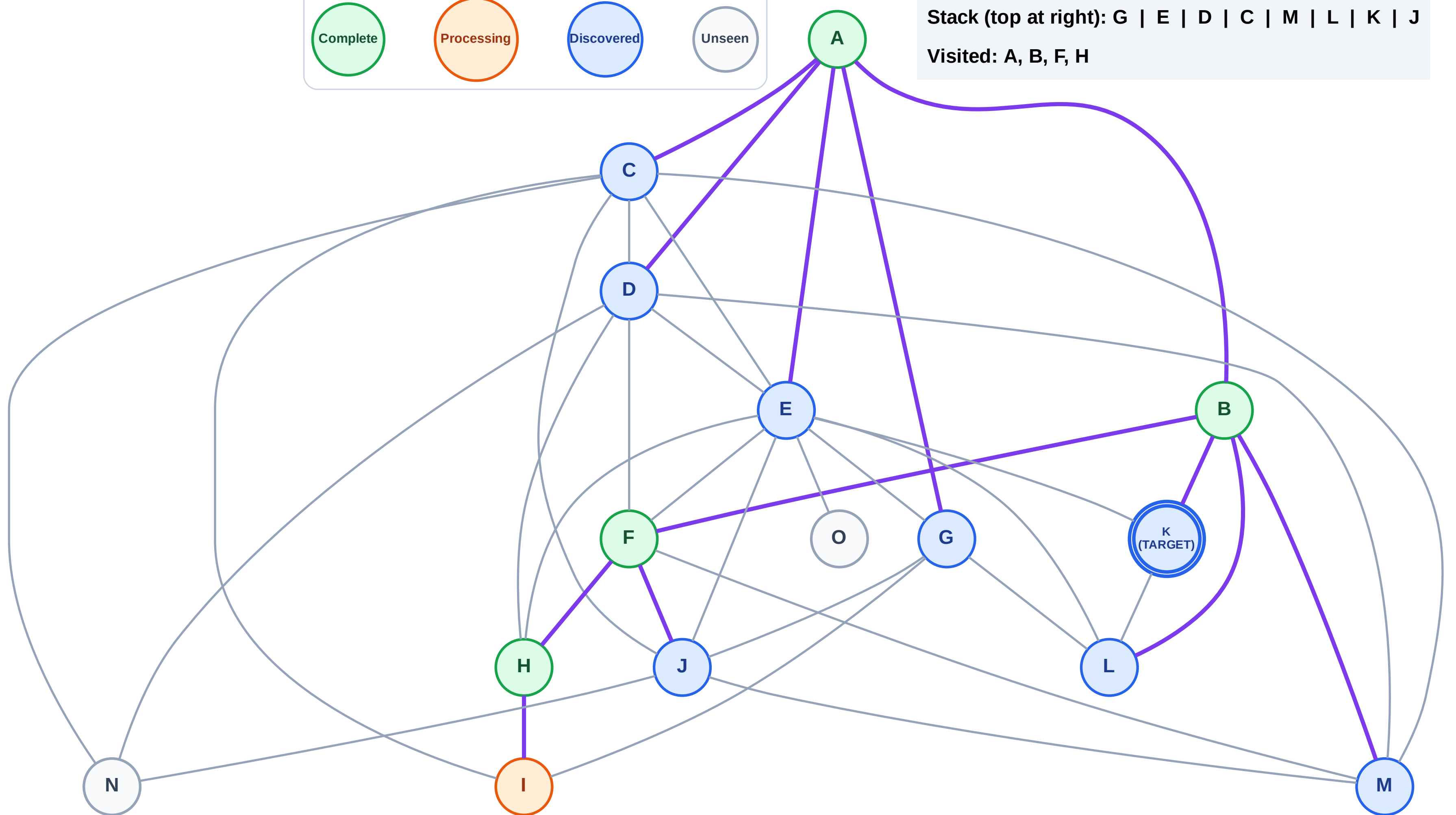
DFS Traversal

Step 10: Process node I

Legend



Stack (top at right): G | E | D | C | M | L | K | J
Visited: A, B, F, H



DFS Traversal

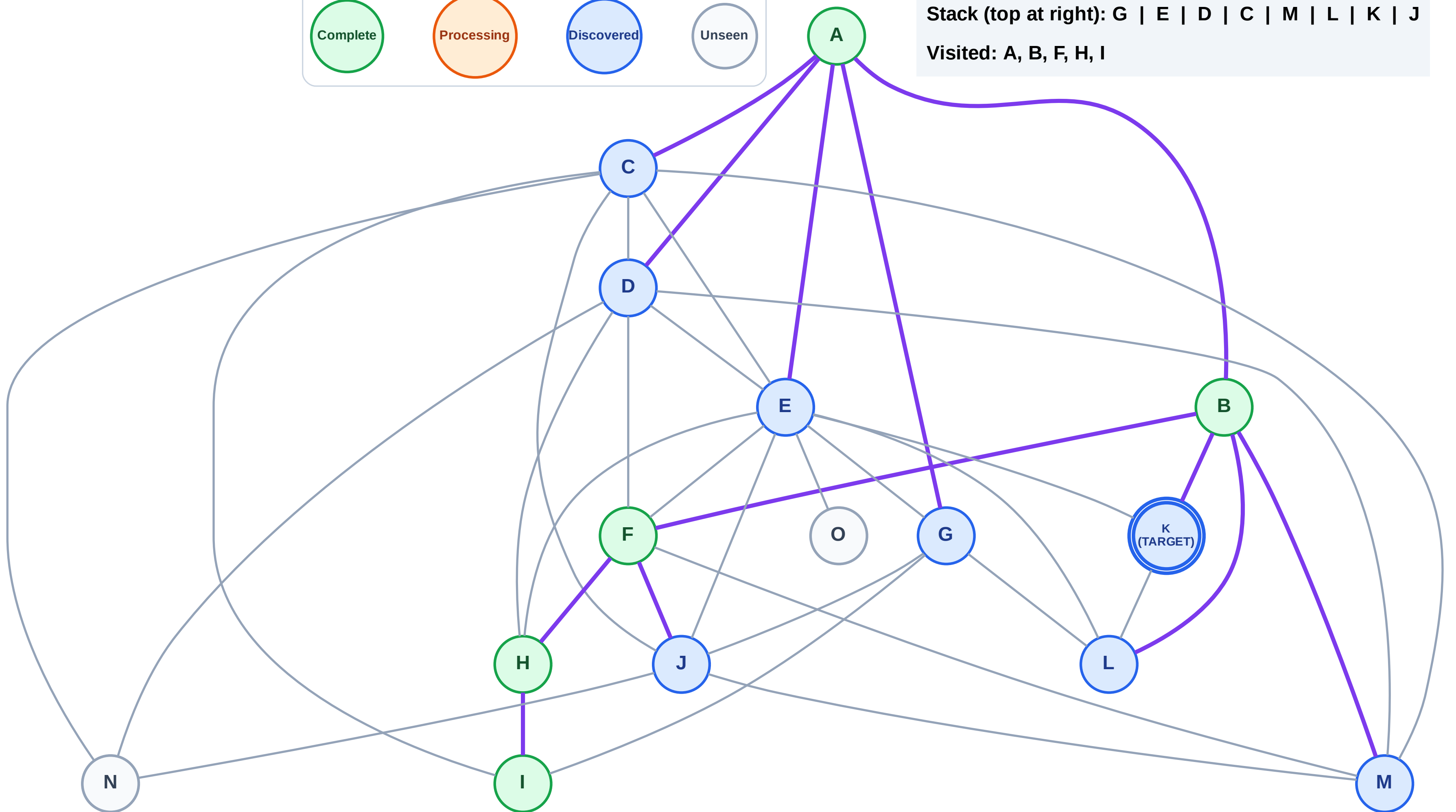
Step 11: No new neighbors from I

Legend



Stack (top at right): G | E | D | C | M | L | K | J

Visited: A, B, F, H, I



DFS Traversal

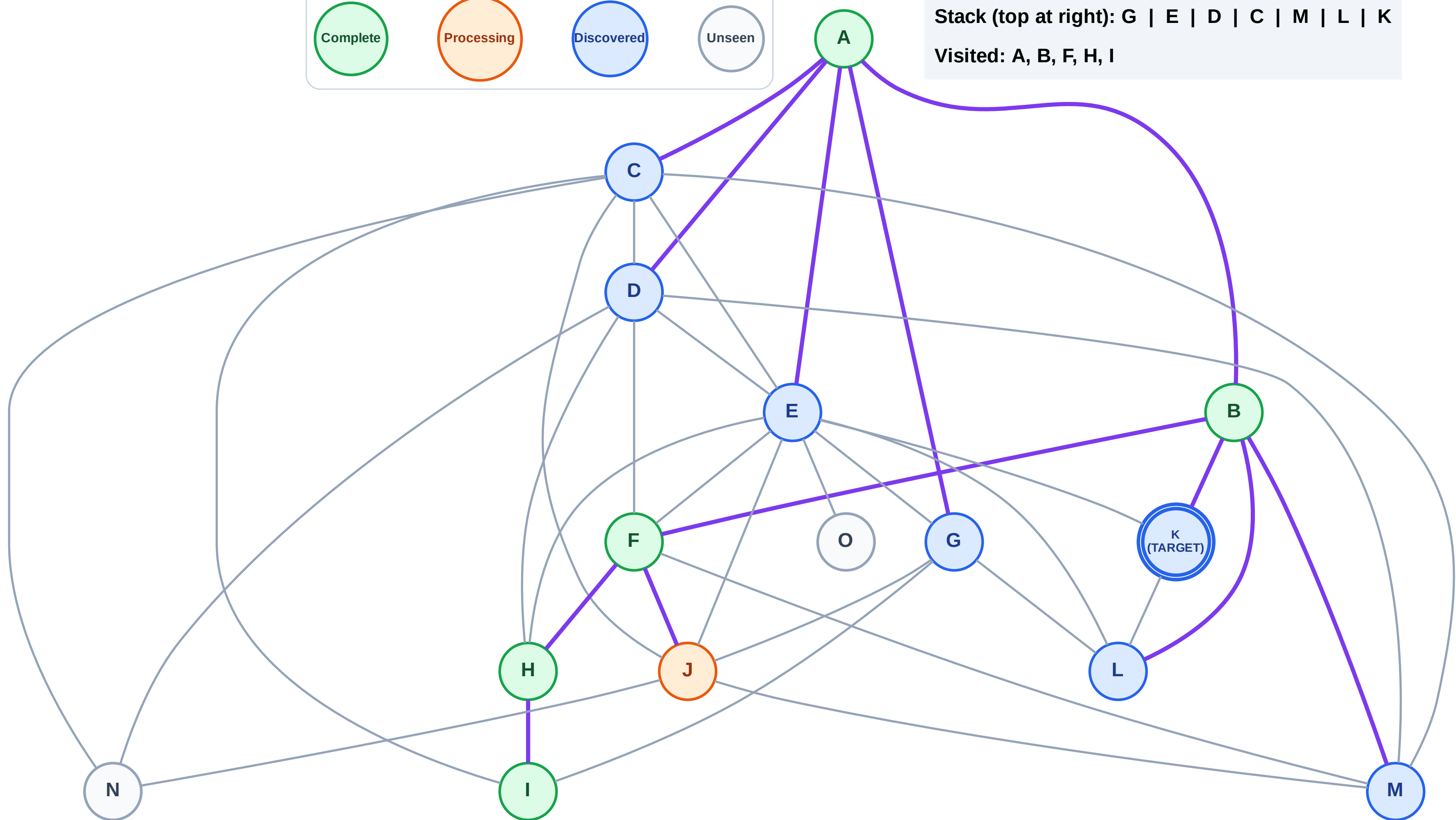
Step 12: Process node J

Legend



Stack (top at right): G | E | D | C | M | L | K

Visited: A, B, F, H, I



DFS Traversal

Step 13: Discover N from J

Legend

Complete

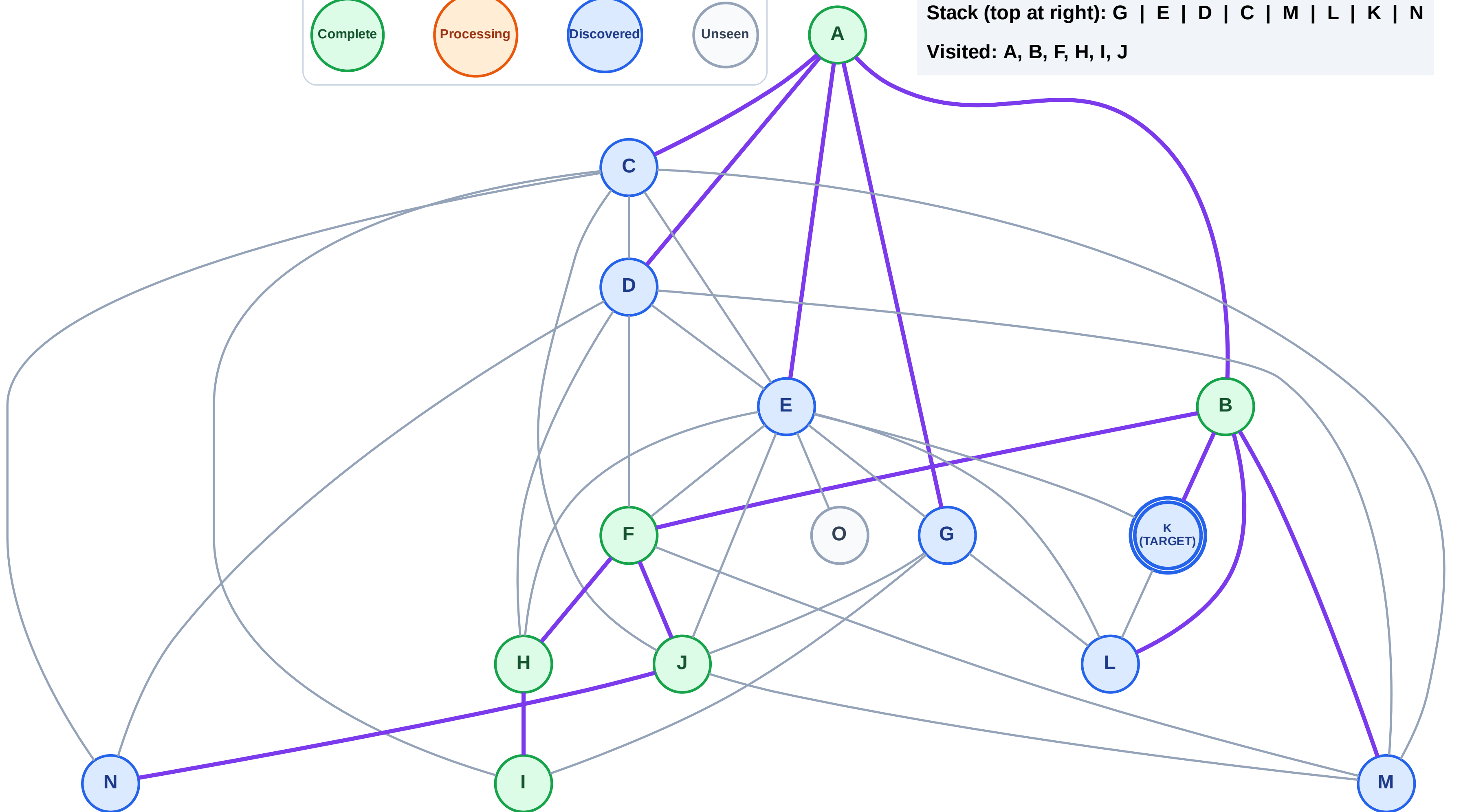
Processing

Discovered

Unseen

Stack (top at right): G | E | D | C | M | L | K | N

Visited: A, B, F, H, I, J



DFS Traversal

Step 14: Process node N

Legend

Complete

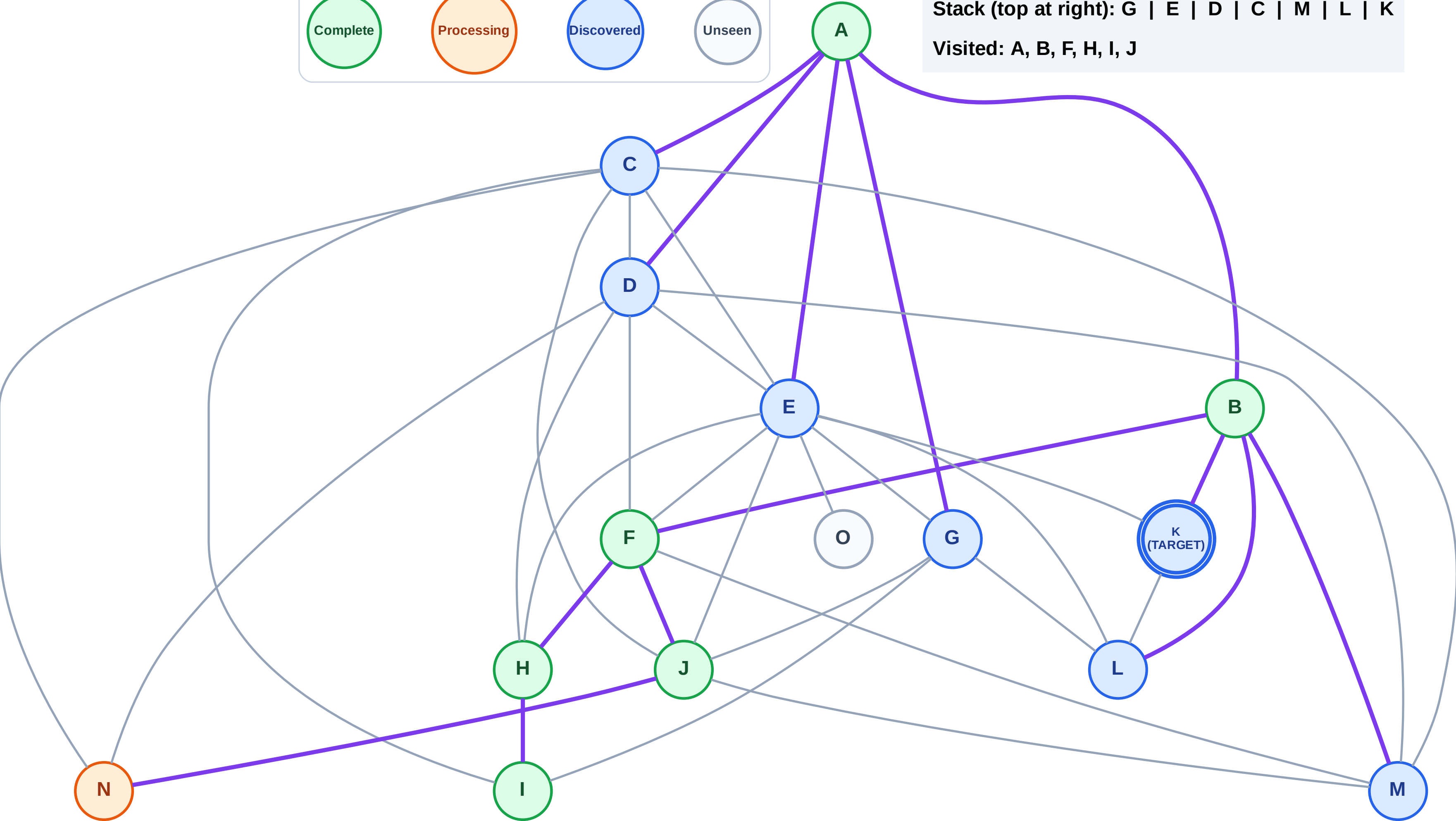
Processing

Discovered

Unseen

Stack (top at right): G | E | D | C | M | L | K

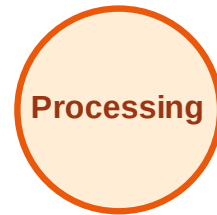
Visited: A, B, F, H, I, J



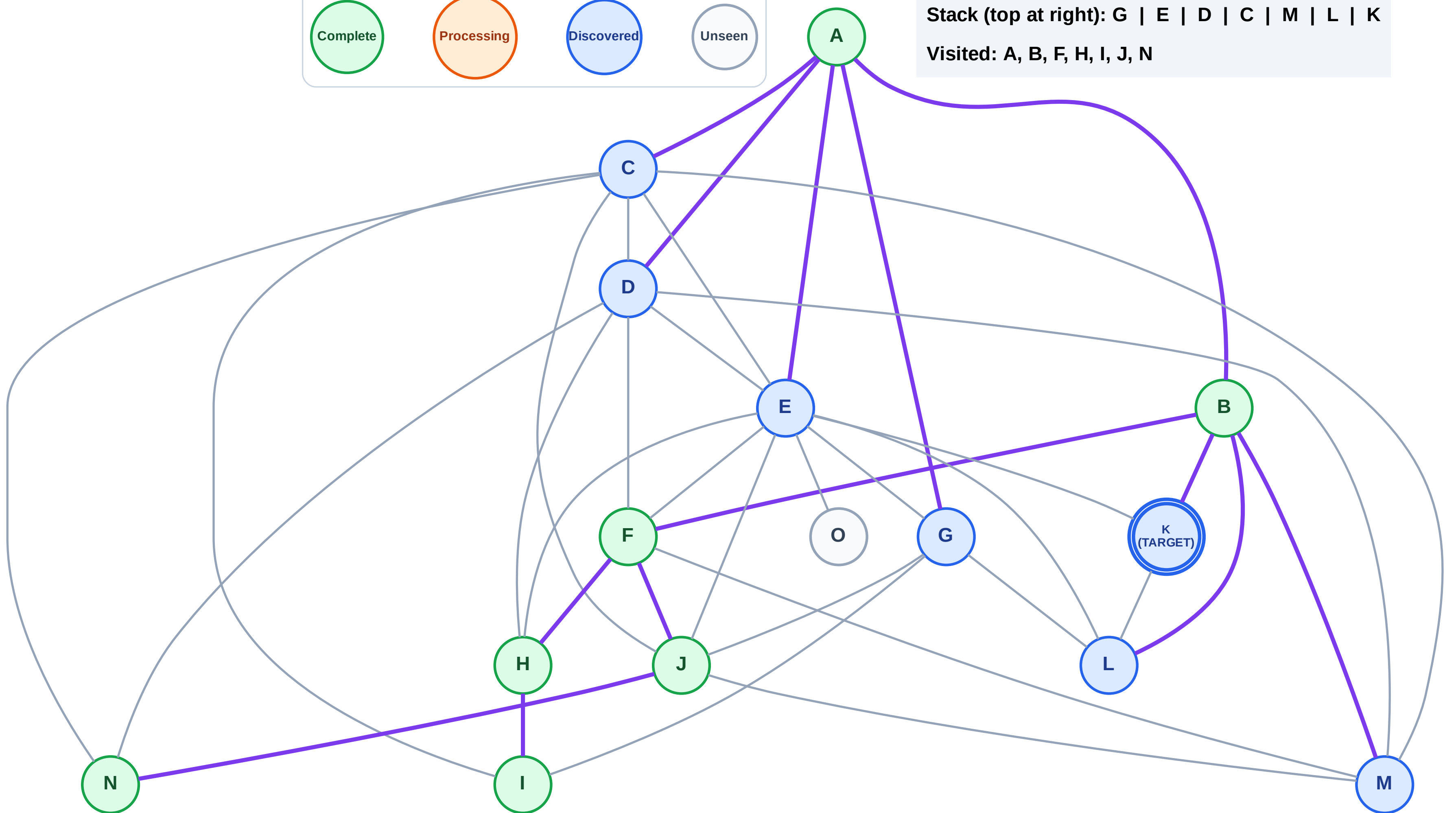
DFS Traversal

Step 15: No new neighbors from N

Legend



Stack (top at right): G | E | D | C | M | L | K
Visited: A, B, F, H, I, J, N



DFS Traversal

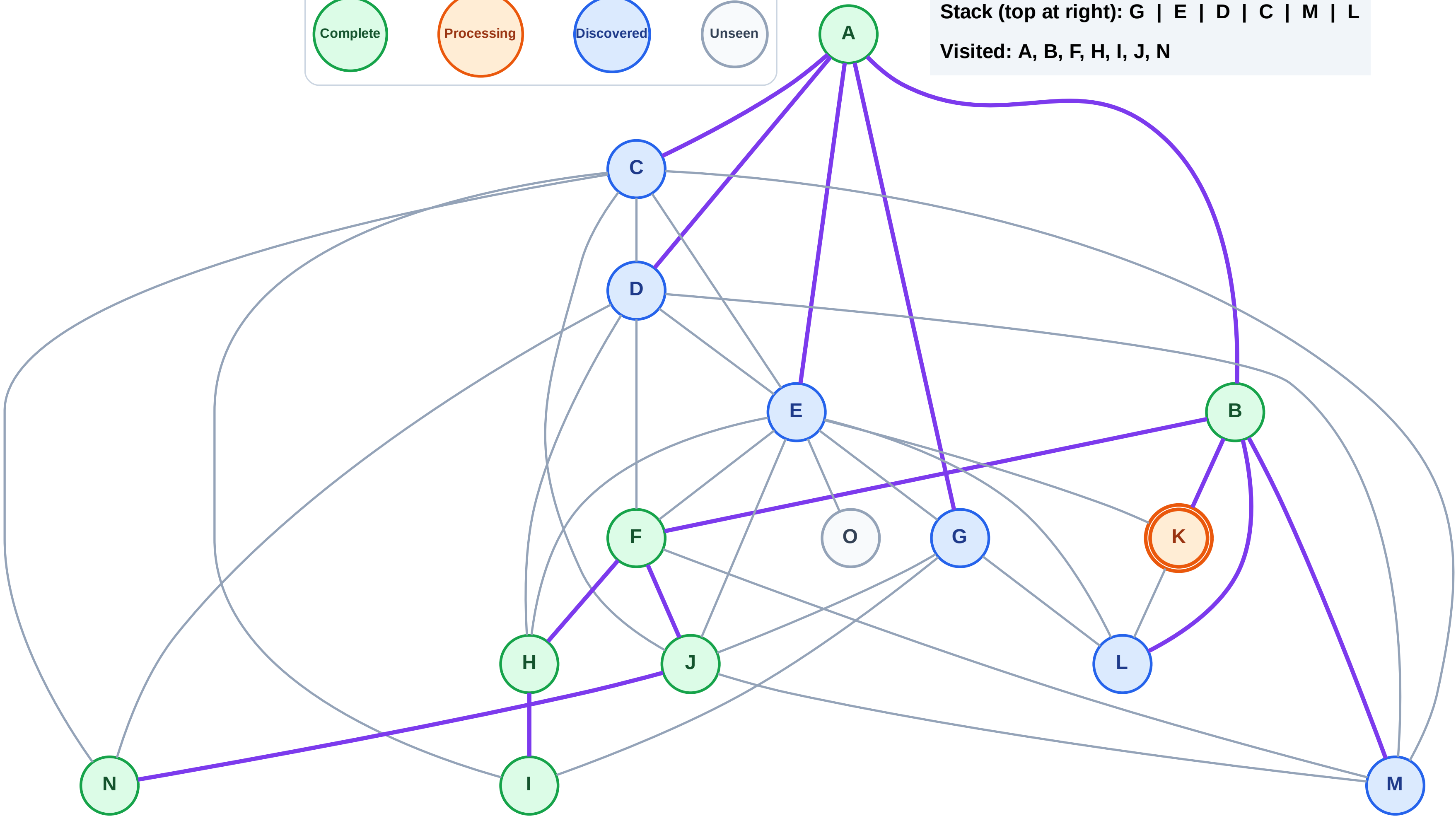
Step 16: Process node K

Legend



Stack (top at right): G | E | D | C | M | L

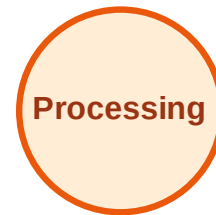
Visited: A, B, F, H, I, J, N



DFS Traversal

Step 17: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend



Stack (top at right): G | E | D | C | M | L

Visited: A, B, F, H, I, J, K, N

